MIL-R-81877B(AS)
10 November 1980
SUPERSEDING
MIL-R-81877A(AS)
22 September, 1975

MILITARY SPECIFICATION

RADIO SETS AN/ARC-159 AND AN/ARC-159(V)

This specification is approved by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide radio frequency reception and transmission of amplitude modulation (AM) signals on any of 7,000 channels in 25 kilohertz (KHz) increments in the 225.000 through 399.975 megahertz (MHz) radio frequency (RF) band. The main function of the equipment is to provide simplex two way communications of:

a. Amplitude Modulation (AM) Normal Voice

b. Amplitude Modulation (AM) Secure Voice

Additionally, the equipment shall provide the capabilities for:

a. Continuous monitoring of the ultrahigh frequency (UHF) AM emergency channel by incorporation of a separate fixed tuned guard receiver.

b. Reception of UHF automatic direction finder (ADF) signals on any selected channel.

c. Transmission of signals from a reference signal generator of the Command Active Sonobuoy System (CASS) or Directional Command Active Sonobuoy System (DICASS) on a preselected channel - AN/ARC-159(V) Systems only.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Specifications and Standards Department, (ESSD) Code 93, Lakehurst, New Jersey 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5821

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the current version before use.

d. Transmission of an internally generated 1,020 hertz (Hz) tone on any selected channel.

e. Automatic relaying whereby interconnection of two radio sets results in automatic transmission on one radio set of either normal or secure voice signals being received by the other radio set.

f. Control of the TSEC/KY-28 secure voice equipment.

g. Remote display of the radio set operating frequency and preset channel number - AN/ARC-159(V) Systems only.

h. Guard channel precedence operation whereby an external control automatically turns the radio set on and tunes it to the guard channel - AN/ARC-159(V) Systems only.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Item	Equipment Designation	Paragraph
Radio Set	AN/ARC-159	3.5.1
Radio Set	AN/ARC-159(V)1	3.5.2
Receiver-Transmitters		3.5.3
Receiver-Transmitter	RT-1150/ARC-159(V)	3.5.3.1
Receiver-Transmitter	RT-1194/ARC-159A(V)	3.5.3.2
Controls, Radio Set		3.5.4
Control, Radio Set	C-9577/ARC-159(V) and C-9577A/ARC-159(V)	3.5.4.1
Control, Radio Set	C-10436/ARC-159(V) and C-10436A/ARC-159(V)	3.5.4.1
Control, Radio Set	C-9815/ARC-159(V)	3.5.4.2
Control, Radio Set	C-9816/ARC-159(V)	3.5.4.2
Control, Radio Set	C-9817/ARC-159(V)	3.5.4.2
Control, Radio Set	C-9818/ARC-159(V)	3.5.4.2

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the current version before use.

Item	Equipment Designation	Paragraph
Control, Radio Set	C-9951/ARC-159(V) and C-9951A/ARC-159(V)	3.5.4.3
Control, Radio Set	C-9952/ARC-159(V) and C-9952A/ARC-159(V)	3.5.4.4
Control, Radio Set	C-9946/ARC-159(V) and C-9946A/ARC-159(V)	3.5.4.5
Control, Radio Set	C-9954/ARC-159(V) and C-9954A/ARC-159(V)	3.5.4.6
Controls, Secure Systems		3.5.5
Control, Radio Set	C-9451/ARC-159(V)	3.5.5.1
Control, Radio Set	C-9960/ARC-159(V)	3.5.5.2
Switching Units		3.5.6
Switching Unit	SA-1964/ARC-159(V)	3.5.6.1
Switching Unit	SA-2078/ARC-159(V)	
Indicator, Frequency-Channel		3.5.7
Indicator, Frequency-Channel	ID-1972/ARC-159(V)	3.5.7.1
Indicator, Frequency-Channel	ID-1984/ARC-159(V)	3.5.7.2
Indicator Group	OD-122/ARC-159(V)	3.5.7.3
Indicator, Frequency-Channel	ID-2053/ARC-159(V)	3.5.7.3.1
Converter, Digital to Digital	CV-3328/ARC-159(V)	3.5.7.3.2
Filter-Converter	F-1420/ARC-159A(V)	3.5.8
Power Supplies		3.5.9
Power Supply - AN/ARA-25	PP-7095/ARC-159(V)	3.5.9.1
Power Supply	PP-7323/ARC-159(V)	3.5.9.2
Cooler, Air, Electronic Equip.	HD-1001/ARC-159A(V)	3.5.10
Mounting Base, Electrical Equip.	MT-4609/ARC-159(V)	3.5.11

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the current version before use.

Item	Equipment Designation	Paragraph
Mounting Base, Electrical Equip.	MT-4658/ARC-159(V)	3.5.12
Mounting Base, Electrical Equip.	MT-4760/ARC-159(V)	3.5.12
Mounting Base, Electrical Equip.	MT-4966/ARC-159(V)	3.5.12
Mounting Base, Electrical Equip.	MT-4967/ARC-159(V)	3.5.12
Mounting Base, Electrical Equip.	MT-4659/ARC-159(V)	3.5.13
Mounting Base, Electrical Equip.	MT-4660/ARC-159(V)	3.5.14
Mounting Base, Electrical Equip.	MT-4661/ARC-159(V) and MT-4661A/ARC-159(V)	3.5.15
Mounting Base, Electrical Equip.	MT-4664/ARC-159(V)	3.5.16
Mounting Base, Electrical Equip.	MT-4795/ARC-159A(V)	3.5.17
Mounting Base, Electrical Equip.	MT-4839/ARC-159A(V)	3.5.18
Mounting Base, Electrical Equip.	MT-4840/ARC-159A(V)	3.5.19
Mounting Base, Electrical Equip.	MT-6093/ARC-159A(V)	3.5.20

1.3 Associated Equipment - The equipment covered by this specification shall operate with the associated equipment listed in 6.8.

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents - The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration
MIL-W-5088	Wiring; Aerospace Vehicle
MIL-E-5400	Electronic Equipment, Airborne, General Specification for

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the current version before use.

Military (Cont'd.)

MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-M-7793	Meter, Time Totalizing
MIL-F-15160	Fuses; Instrument, Power and Telephone
MIL-E-17555	Electronic and Electrical Equipment, Accessories and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical and Aeronautical Support Equipment
MIL-T-23103	Thermal Performance Evaluation, Airborne Electronic Equipment; General Requirement for
MIL-C-25050	Color, Aeronautical Lights, and Lighting Equipment, Specification for
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-I-81219	Indicator, Elapsed Time, Electrochemical
MIL-M-81288	Mounting Bases, Flexible Plastic Foam
MIL-G-81665	Generator Transmitter Group AN/ASA-76

Naval Air Systems Command

AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for

STANDARDS

Federal

FED-STD-595	Colors

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

Military

MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Aircraft Electric Power Requirements
MIL-STD-781	Reliability Tests; Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-882	Subsystem Safety Program; Requirements for
MIL-STD-965	Parts Control Program
MIL-STD-2074	Failure Classification for Reliability Testing
MIL-STD-2076	Unit Under Test Compatibility with Automatic Test Equipment, General Requirements for
MIL-STD-2077	Test Program Sets, General Requirements for

PUBLICATIONS

Naval Air Systems Command

EI-603	Avionics Installation Instruction for Radio Sets AN/ARC-159 and AN/ARC-159(V)
ET-603	Avionics Bench, Preflight and Flight Test Procedures for Radio Sets AN/ARC-159 and AN/ARC-159(V)

National Security Agency

CSEEB-13	TSEC/KY-28 Speech Security Equipment Engineering Bulletin
KAM-213/TSEC	TSEC/KY-28 Speech Security Equipment Limited Maintenance Manual

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T04:37Z
Check the source to verify that this is the currentversion before use.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First Article - When specified (see 6.2), the contractor shall furnish a sample unit for first article inspection and approval (see 4.2 and 6.2.1).

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Microelectronic Technology shall be considered and microelectronic items shall conform to requirements specified herein.

b. Other parts and materials requirements shall conform to MIL-E-5400.

c. Nonrepairable subassemblies shall be used in accordance with AR-10 and as outlined in MIL-E-5400.

d. When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of MIL-STD-965.

3.2.2 Modules - The electronic portions of the equipment shall be functionally modularized in accordance with AR-10. Conformal coatings, encapsulants, embedments or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.3 Design and Construction - The equipment shall meet the requirements of MIL-E-5400 for design construction, and workmanship, as specified herein. The design shall recognize equipment safety requirements and include implementation in accordance with the plan of MIL-STD-882.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.3.1 Total Weight - The total weight of the radio sets shall be a minimum consistent with good design. The maximum weights of the individual equipments, excluding external connectors and cables, shall be not greater than the limits specified in 3.5.

3.3.2 Reliability - The contractor shall conduct a reliability program in accordance with MIL-STD-785. On a reorder from a contractor who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract. (see 6.2.1)

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently, for a period of at least 1,000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 10,000 hours with standard military servicing and replacement of parts.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The following equipment shall meet the time requirement specified below in mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.2. The specific mean time between failures of the respective equipments, systems, and ancillary units shall be as follows:

a.	AN/ARC-159	1000 hours
b.	AN/ARC-159(V)1	1000 hours
c.	AN/ARC-159(V)2	1000 hours
d.	AN/ARC-159(V)3	1000 hours
e.	AN/ARC-159(V)4	1000 hours
f.	AN/ARC-159(V)5	1000 hours
g.	AN/ARC-159(V)8	1000 hours
h.	AN/ARC-159A(V)5 (less F-1420/ARC-159A(V))	1000 hours
i.	F-1420/ARC-159A(V)	4000 hours
j.	Indicator Channel Frequency	8000 hours
k.	Switching Unit and Associated Control	8000 hours
l.	Power Supply - Converter	8000 hours
m.	Power Supply - PP-7095/ARC-159(V)	8000 hours

Radio Set Control Units (when tested individually) 5000 hours. One control unit shall normally be considered as part of any system requiring a control unit to operate a receiver-transmitter. Combinations of basic equipments and ancillary units shall meet the MTBF calculated for the system from the individual unit MTBF requirements. For AN/ARC-159A(V) system procurements which include the HD-1001/ARC-159A(V), this unit shall be considered as part of the system for reliability testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of wiring and connectors in accordance with MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. Refer to EI-603 for interconnection wiring diagrams.

3.3.4 Control Panels - All rack or console mounted radio sets and control panels shall conform to the requirements of MIL-C-6781 (Type I) to the extent specified herein.

3.3.4.1 Lighting of Console Mounted Equipment - All console mounted control panels and radio sets shall have integrally illuminated type V lighting in accordance with MIL-P-7788, with each having the option of using either class I-R (Navy Red) or class I-W (Air Force white in accordance with MIL-L-27160) color lighting. The C-10436/ARC-159(V) and C-10436A/ARC-159(V) lighting color shall be IPL white in accordance with MIL-C-25050 instead of MIL-P-7788 or MIL-L-27160. Separate input power pins shall be provided for 5 volt (AC or DC) and 28 volt (AC or DC) such that application of power to the assigned pin provides proper illumination. When AC power is applied to the panel lamps, the 600 ohm receiver audio output (noise) under no signal conditions shall not increase more than 12 millivolts.

3.3.4.2 Marking of Console Mounted Equipment - The plastic plate for console mounted control panels and radio sets shall be marked in accordance with MIL-C-6781 or as approved by the procuring activity. The function of all controls shall be clearly identified using approved lighting, letter size and contrasting colors such that each and every function word can be read easily from a distance of 24 inches. Control position identifications shall be clearly and permanently placed on the plastic plate as close as practical to the control position.

3.3.4.3 Console Mounted Equipment Function Marking - The indicators, console mounted control panels and radio sets shall be permanently and clearly marked to indicate the function of the equipment they are associated with. Indicators shall be identified by the abbreviation UHF centrally located on the front panel. Console mounted radio sets and control panels (except C-9451/ARC-159(V)) shall be identified by the abbreviation UHF placed vertically on the left hand side of the control panel. The C-9451/ARC-159(V) shall be identified by the abbreviation KY-28 on the right hand side of the control panel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.3.4.4 Control Knobs and Functions - The number of controls required to operate the equipment shall be held to a minimum consistent with performing all the functions required by this specification. Closely associated controls, such as control knobs on concentric shafts, shall be easily identified by contrasting colors of black and grey.

3.3.4.5 Mounting of Console Mounted Equipment - The console mounted control panels and radio sets shall be securely mounted in an aircraft console or instrument panel using quick disconnect fasteners in accordance with MIL-C-6781. Under these mounting conditions, they shall be considered hard mounted and meet all requirements of this specification when mounted in any attitude. When mounted in a console or instrument panel, the front panel shall not extend out from the outside surface of the mounting rails more than 0.50 inches and the control knobs shall not extend out more than 1.75 inches from the surface of the front panel. The center of gravity of the equipments shall be on or near their centerlines as shown in EI-603.

3.3.4.6 Watertightness of Console Mounted Equipments - The portion of the console mounted radio sets and control panels (except C-9451/ARC-159(V) and C-9960/ARC-159(V) in front of the console mounting surface shall be watertight in accordance with MIL-STD-810. Water condensation may occur within the equipment; however, performance shall not be degraded due to the water condensation. (See 4.2.2)

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of MIL-E-5400. The equipment shall be constructed to preclude the requirement for any adjustments when replacing an equipment by another equipment of the same type designation.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-STD-461, except as modified below:

a. CE01 Conducted Emission 30 Hz to 20 KHz

Broadband emissions on the +28 volt power lead in the frequency range of 30 Hz to 20 KHz may exceed the specified level of the 90 dB above 1 μa/20 KHz by 15 dB.

b. CE06 Conducted Emission 10 KHz to 12.4 GHz

Receive Mode
The broadband emissions from the antenna terminal in the 15 KHz to 25 KHz range, the 1.1 MHz to 1.4 MHz range and the 4.0 MHz to 8.5 MHz range may exceed the specified maximum levels by 4 dB. The narrowband emissions at the third harmonic of the first injection frequency may exceed the specified levels by 10 dB.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the currentversion before use.

b. CE06 Conducted Emission 10 KHz to 12.4 GHz (Cont'd.)

Transmit Mode
The emission levels at the synthesizer frequency and the fundamental harmonic frequencies may exceed the specified limits by 6 dB.

c. CS01 Conducted Susceptibility 30 Hz to 50 KHz

The maximum level of the interfering signal at frequencies from 30 Hz to 300 Hz shall be 1.0 volts instead of 2.8 volts.

d. CS02 Conducted Susceptibility 50 KHz to 400 MHz

The maximum level of interfering signal at the receiver channel frequency, for normal receiver performance, shall be 2 millivolts instead of 1 volt. The interfering signal level at the receiver channel image frequencies shall be 20 millivolts maximum.

e. CS03 Conducted Susceptibility 30 Hz to 10 GHz

The receiver immunity to intermodulation products may be less than the specified limit of 66 dB by up to 13 dB.

f. RE02 Radiated Emission 14 KHz to 10 GHz

Radiated Emission in the receive mode in the range of 100 MHz to 400 MHz may exceed the specified limits by 9 dB. Radiated emissions at the third harmonic of the first injection frequency may exceed the specified limits by 12 dB. The radiated emission limit shall not apply for the fundamental frequency when in the transmit mode.

g. RS03 Radiated Susceptibility 14 KHz to 10 GHz

The interfering signal level at the main receiver channel frequency shall be 3 mv/meter, maximum. The maximum level at the guard receiver channel frequency shall be 600 μv/meter. Interfering signal maximum levels at other frequencies shall be:

	Frequency	Level
1.	241 MHz	2 mv/meter
2.	500 to 1000 MHz	12 mv/meter
3.	1000 to 2000 MHz	200 mv/meter
4.	Image Frequency	400 mv/meter

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the currentversion before use.

3.3.7 Maintainability - The equipment shall have the following maintainability characteristics.

3.3.7.1 Compatibility with VAST - The equipment is not required to meet the automatic test compatibility parameters of MIL-STD-2076, MIL-STD-2077, and AR-10. WRA connectors for test purposes and built-in test (BIT) functions for VAST (AR-10 paragraphs 3.1.4.1, 3.3 and 3.5.4.2) are not applicable.

3.3.7.2 Provision for Maintainability - The maintainability provisions, construction, packaging, test points, and fault isolation characteristics shall be as specified below:

a. Modularization

The equipment shall be of modular construction with modules designed for repairability by replacement of defective sub-assemblies and/or piece parts.

b. Shop Non-Ambiguity (SN-A) Ratio

1. In at least 85 percent of the cases of probable malfuction of an SRA, the fault shall be isolated to the sole SRA.

2. In 90 percent or more, of the cases of probable malfunction of an SRA, the fault shall be isolated to the SRA, and no more than one other SRA.

3. In all cases of probable malfunction of an SRA, the fault shall be isolated to that SRA and no more than two other SRA's.

The above is applicable to all panel mounted radio and receiver-transmitters. Other AN/ARC-159 and AN/ARC-159(V) WRA units shall be exempt from the SN-A ratio requirements.

c. Test Point Provisions

The panel mounted radios and receiver-transmitter units shall have test points adequate to provide a straight forward end to end performance check and to isolate faults to the replaceable module level. Test points shall be located in test point fields under access covers. At the organizational test level, the test points shall enable determination whether or not the equipment is performing satisfactorily and to ascertain which WRA contains the fault. Operational cables may be disconnected during test and fault isolation procedures. The test point requirements of 3.5.4.2 of AR-10 for shop repair are not applicable to the AN/ARC-159 units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.3.8 Nomenclature, Nameplates and Identification - Marking - Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with MIL-N-18307 except for 3.5.7 units where size precludes the use of standard nameplates. The plates may be configured to the space available.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

a.	Temperature	Room ambient (25°C ± 10°C)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input power voltage	27.5 ± 0.5 VDC

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class I equipment, except as modified herein.

3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves III (without isolators) and IV (with isolators) of MIL-E-5400. The console mounted control panels, radio sets, and indicators shall meet Curve III requirements without rear supports.

3.3.11 Warm-Up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall be not greater than the limits outlined in 3.5.

3.3.12 Input Electrical Power -

3.3.12.1 Operating Power - The equipment shall operate as specified herein when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall be 28 volts, DC power, Category B and 115 volts 400 Hz Category B power and shall be not greater than the limits for the respective equipments listed below.

Equipment Designation	28 Volt DC Power Maximum Current	115V AC Power Volt-Ampere
a. AN/ARC-159		
Transmit	4.25 amperes	
Receive or Standby	1.4 amperes	
b. AN/ARC-159(V)1		
Transmit	4.45 amperes	
Receive or Standby	1.6 amperes	
c. RT-1150/ARC-159(V)		
Transmit	4.05 amperes	
Receive or Standby	1.3 amperes	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

Equipment Designation	28 Volt DC Power Maximum Current	115V AC Power Volt-Ampere
d. RT-1194A(V)/ARC-159		
Transmit	12 amperes	
Receive or Standby	2.0 amperes	
e. C-9577/ARC-159(V)	0.8 amperes	
C-9577A/ARC-159(V)	0.8 amperes	
C-10436/ARC-159(V)	0.8 amperes	
C-10436A/ARC-159(V)	0.8 amperes	
f. C-9815/ARC-159(V)	0.9 amperes	
g. C-9816/ARC-159(V)	0.9 amperes	
h. C-9817/ARC-159(V)	0.9 amperes	
i. C-9818/ARC-159(V)	0.9 amperes	
j. C-9951/ARC-159(V)	0.9 amperes	
C-9951A/ARC-159(V)	0.9 amperes	
k. C-9952/ARC-159(V)	0.9 amperes	
C-9952A/ARC-159(V)	0.9 amperes	
l. C-9946/ARC-159(V)	0.9 amperes	
C-9946A/ARC-159(V)	0.9 amperes	
m. C-9954/ARC-159(V)	0.9 amperes	
C-9954A/ARC-159(V)	0.9 amperes	
n. C-9451/ARC-159(V)	0.2 amperes	
o. C-9960/ARC-159(V)	0.4 amperes	
p. SA-1964/ARC-159(V)	0.5 amperes	
q. SA-2078/ARC-159(V)	0.6 amperes	
r. ID-1972/ARC-159(V)	0.7 amperes	
s. ID-1984/ARC-159(V)	0.7 amperes	
t. OD-122/ARC-159(V)	0.7 amperes	
u. F-1420/ARC-159(V)	1.6 amperes	
v. PP-7095/ARC-159(V)	0.8 amperes	
w. PP-7323/ARC-159(V)		140 VA

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

Equipment Designation	28 Volt DC Power Maximum Current	115V AC Power Volt-Ampere
x. HD-1001/ARC-159(V)		28 VA
y. MT-4659/ARC-159(V)	0.2 amperes	
z. MT-4660/ARC-159(V)	0.2 amperes	
aa. MT-4661/ARC-159(V)	0.2 amperes	
MT-4661A/ARC-159(V)	0.2 amperes	28 VA
ab. MT-4664/ARC-159(V)	0.2 amperes	
ac. MT-4795/ARC-159(V)	0.2 amperes	
ad. MT-4966/ARC-159(V)		28 VA
ae. MT-4967/ARC-159(V)		28 VA
af. MT-6093/ARC-159(V)	0.3 amperes	

3.3.12.2 Lighting Power - Input power for lighting is not included in the limits of 3.3.12.1, but shall be not greater than 5 watts for any individual red lighted unit or 5.5 watts for any white lighted unit.

3.3.12.3 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.3.13 Cooling -

3.3.13.1 Thermal Conditions - The equipment, except for the RT-1194/ARC-159A(V), shall operate satisfactorily under the thermal conditions described by MIL-E-5400, Class I, without the requirement for auxiliary cooling. The RT-1194/ARC-159A(V) shall be supplied with cooling air as defined in 3.5.3.2.5. Thermal performance analysis requirements of MIL-T-23103 shall be satisfied by successful completion of temperature-altitude testing during preproduction tests.

3.3.14 Time Totalizing Meter - The following equipments shall contain a 5,000 hour scale time totalizing meter in accordance with MIL-I-81219.

Equipment Designation	Type of Meter
AN/ARC-159	MS 3311-2
AN/ARC-159(V)1	MS 3311-2
RT-1150/ARC-159(V)	MS 3311-2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

The following equipment shall contain a 9999 hour scale time totalizing meter in accordance with MIL-M-7793.

Equipment Designation

RT-1194/ARC-159A(V)

3.4 Performance - Unless otherwise specified herein, values set forth to establish the requirements of satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptance variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The equipment shall provide the functions and capabilities listed in 1.1 for operation in military aircraft and helicopters. Additionally, the equipment shall be compatible with equipment listed in 6.7 to provide operations outlined in 6.1 and 3.5.

3.5 Detail Requirements -

3.5.1 Radio Set AN/ARC-159 - The radio set shall meet the following requirements:

3.5.1.1 Function - The radio set shall provide simplex two way communication of AM voice (Normal and Secure) on any of 7000 channels in 25 KHz increments in the 225.000 through 399.975 MHz RF Band. Additionally, the radio set shall provide the capabilities for:

a. Continuous monitoring of the UHF AM emergency RF channel (243.000 MHz) by incorporation of a separate fixed tuned guard receiver, with provisions for changing to other fixed frequencies between 238.000 and 248.000 MHz.

b. Reception of UHF ADF signals on any selected channel.

c. Transmission of an internally generated 1020 Hz tone on any selected channel.

d. Transmission and reception of normal or secure voice signals for automatic relaying when connected with a like radio set AN/ARC-159 or AN/ARC-159(V), radio set control C-9451/ARC-159(V) and switching unit SA-1964/ARC-159(V) or radio set control C-9960/ARC-159(V) and switching unit SA-2078/ARC-159(V).

e. Display of the radio frequency or preset channel number selected for operation.

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.1.2 Form Factor - The radio set shall be not greater than 9.25 inches by 5.75 inches by 4.875 inches (including control knobs and integral connectors) as shown in EI-603.

3.5.1.3 Weight - The weight of the radio set shall be not greater than 8.75 pounds.

3.5.1.4 Contents - The radio set shall be one self-contained unit of all solid state modular electronic circuit design containing the following modules and assemblies:

a. Synthesizer Module -
b. RF Translator/Modulator Module -
c. IF/Audio Amplifier Module -
d. Guard Receiver Module -
e. Power Amplifier/Power Supply Module -
f. Control/Indicator Assembly -

3.5.1.5 Controls - Controls required for operation of the radio set shall be located on the Control/Indicator Assembly at the front of the radio set. The number of controls required to operate the radio set shall be held to a minimum consistent with performing all the functions defined in this specification.

3.5.1.5.1 Function Selector Switch - This switch shall be a four position switch allowing the operator to select any one of the functions specified below:

a. OFF Function - In this switch position, identified by "OFF", the DC input power shall be removed from the radio set. The switch shall operate in the plus (+) DC input power lead. The radio set shall contain a reverse polarity protection circuit to prevent damage to any part of the radio set by applying proper input power voltage level of reverse polarity.

b. Main Receiver-Transmitter Function - In this switch position, identified by "MAIN", the main receiver shall meet all the performance capabilities defined herein. The guard receiver shall be in the off condition. The transmitter shall be in the standby condition. Also in this switch position, the radio set shall be capable of transmitting and meeting all the performance capabilities defined in this specification when the external transmit/receive switch is closed. In the transmit condition, the main receiver shall be turned off and the RF output of the radio set shall be switched to the transmitter output terminals instead of receiver input terminals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

c. Main Receiver-Transmitter Plus Guard Function - In this switch position, identified by "BOTH", both main and guard receivers shall be turned on and the operator shall be able to hear the audio output of both receivers simultaneously. The radio set shall have the same transmit-receive capabilities as defined in the main receiver-transmitter function switch position with the addition that in the transmit condition both the main and guard receivers shall be turned off and only the transmitter output connected to RF output terminals of the radio set.

d. Automatic Direction Finder Function - In this switch position, identified by "ADF", the main receiver-transmitter shall be operating as defined in 3.5.1.5.1 (c) above, plus the radio set shall be capable of operation with the associated automatic direction finder equipment listed in 6.7. In this switch position the guard receiver shall remain on.

3.5.1.5.2 Frequency Selector Mode Switch - The frequency selector mode switch shall be a four-position switch allowing the operator to select guard, manual, preset and preset read modes of frequency selection.

a. Guard Mode - This switch position, identified by the word "GUARD", shall automatically tune the main receiver and transmitter to the guard channel (normally 243.000 MHz). The guard frequency shall be shown on the front panel using the display described in 3.5.1.8. When operating in this mode, the preset frequency selector control as well as the manual frequency selector controls shall be in-operative.

b. Manual Mode - This switch position, identified by the word "MANUAL", shall allow the operator to rapidly and easily select any one of the possible 7,000 operating frequencies by use of the controls described in 3.5.1.5.8. Any operating channel frequency selected shall be shown on the front panel using the display described in 3.5.1.8. When operating in the manual mode, it shall be impossible to change or affect the operating frequency of the radio set by means of the preset frequency select control.

c. Preset Mode - This switch position, identified by the word "PRESET", shall allow the operator to rapidly and easily select by one control, any one of the possible 20 preset operating channels by use of the control described in 3.5.1.5.9. The channel number (1 through 20) of any operating channel selected shall

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:37Z
Check the source to verify that this is the current version before use.

be shown on the display described in 3.5.1.8. The preset channel frequency may be written on the front panel using the frequency chart described in 3.5.1.7. When operating in this mode, it shall be impossible to change the operating frequency of the radio set by means of the manual frequency selector controls.

d. Preset Read Mode - This switch position, identified by the word "READ", shall display the preset channel frequency to which the transceiver is tuned in lieu of the preset channel number. The frequency shall be shown on the display described in 3.5.1.8.

3.5.1.5.3 Tone Transmit Switch - This spring-return push-button switch, identified by the word "TONE", shall allow the operator to transmit an internally generated 1020 Hz tone modulated signal when the switch is depressed. When transmitting a tone signal, both the main and guard receivers shall be turned off and the RF output of the radio set shall be switched to the transmitter output terminals instead of the receiver input terminals.

3.5.1.5.4 Audio Volume Control - The audio volume control, identified by the word "VOL", shall allow the operator to manually adjust the audio power output level of 3.5.1.17.13 as desired. This control shall provide rated (250 milliwatt under standard conditions) audio output in the full clockwise position with attenuation increasing at least 40 dB when rotating in the counter-clockwise direction but prior to reaching full counter-clockwise position. In full counter-clockwise position, the audio output shall be turned off. The attenuation change, in dB, shall be essentially linear with the degree of rotation. This control shall adjust the audio output level of the main receiver, the guard receiver, the TSEC/KY-28 deciphered receive audio (external wiring required), the transmitter sidetone, and the TSEC/KY-28 transmit sidetone (external wiring required). Refer to EI-603 for external wiring required.

3.5.1.5.5 Squelch Disable Switch - The radio shall have a switch located on the front panel that shall allow the operator to disable the squelch circuit on the main receiver only. The switch shall be identified at squelch with the on and off conditions clearly identified. In the on condition the squelch circuit shall be enabled. In the off condition the squelch circuit shall be disabled to allow receiving of main receiver signals regardless of carrier to noise characteristics.

3.5.1.5.6 Light Dimmer Control - A rotary light dimmer control shall be provided on the front panel to adjust the illumination intensity of the incandescent lamps of the Frequency/Channel Display by varying their input voltage downward from the maximum rated voltage.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the currentversion before use.

3.5.1.5.7 Lamp Test Switch - A switch shall be provided on the front panel to test the incandescent lamps of the Frequency/Channel Display. Activation of this switch shall illuminate all the elements of the seven segment units in all six digits.

3.5.1.5.8 Manual Frequency Select Controls - Four separate detented selector controls shall be provided on the front panel for use in manual selection of any of the 7000 available channels. These controls shall select the following respective frequency increments: 100 MHz and 10 MHz, 1 MHz, 0.1 MHz, and 25 kHz.

3.5.1.5.9 Preset Frequency Select Control - One detented selector control shall be provided on the front panel for selection of any one of the 20 available preset channels.

3.5.1.5.10 Main and Guard Receiver Squelch Level Controls - Manual screwdriver adjusted squelch level controls shall allow the operator to adjust and set the RF level at which the main and guard receiver squelch operates. These squelch level controls shall be located behind a hinged access door on the radio front panel and shall be marked as follows:

SQUELCH

GUARD MAIN

3.5.1.6 Preset Channel Memory Device - A 20 channel preset memory device shall be provided in the Control/Indicator Assembly. With this device it shall be possible to select any 20 of the 7000 channel frequencies by using the control described in 3.5.1.5.9. The memory device shall be a non-destruct type such that power line interruptions or any comination of environmental conditions will not cause a loss of memory. Any special tool required for internally resetting this device to result in different preset channel frequencies shall be attached and available to the operator.

3.5.1.7 Frequency Chart - A frequency chart shall be provided on the front of the radio set to allow the operator to record in pencil the channel frequency for each, and all, of the twenty preset channels. The surface of the chart shall be such that pencil entries may be clearly recorded and erased for as many as 1,000 times. Each channel frequency shall be defined with respect to the preset channel number. The chart shall be physically located on the outside surface of the hinged access door to the preset channel memory device and easily visible to the operator when the access door is closed.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the currentversion before use.

3.5.1.8 Frequency/Channel Display - A six digit frequency/channel display visible through a red filter shall be provided on the front of the radio set to display the operating radio frequency or preset channel number. Seven-segment tungsten filament incandescent lamps shall be used for each digit of the display. For radio frequency display (applicable for GUARD, MANUAL and PRESET READ positions of the frequency mode selector switch) a decimal shall also be lighted to identify the MHz position to three significant digits (i.e. 225.000). The abbreviation MHz shall not be printed beside or on the display. For PRESET channel number display, only one or two digits (for PRESET channel numbers 1 through 20) shall be lighted with the four remaining digits unlighted. The display shall be arranged horizontally in a centrally located position on the front of the radio set. The size of each digit of the display shall be such that the digits can be easily read from a distance of 2.0 feet at any angle up to at least 30° from an equipment centerline perpendicular to the surface of the front panel. The brightness, color, shape and control of the electronic readouts shall be subject to procuring activity approval. Provisions shall be made for easy removal and replacement of the incandescent sevel element frequency display lamps without loosening or removing the equipment front panel or control knobs.

3.5.1.9 Duty Cycle - The radio set shall have at least a one minute transmit - five minute receive duty cycle. For extended periods of transmit operation, thermal sensing circuits in the transmitter shall automatically reduce the RF power output level to maintain safe operating temperatures and prevent damage to the unit. The radio set shall contain a thermal circuit for protection against the overheating of the transmitter power amplifiers. If, by inadvertently leaving the radio set in the transmit condition or some other means, the transmitter power amplifiers become overheated, the thermal protective circuit shall so reduce the drive power to the power amplifiers that no damage will occur to any component part and the radio set will satisfactorily operate again when the temperature of the power amplifier is returned to normal. The RF power output under any required combination of extreme service conditions, voltage standing wave radio (VSWR) and extended time transmission shall be 1 watt minimum.

3.5.1.10 Emergency Power - During operation in the emergency steady state electric power category B condition as defined in MIL-STD-704, the equipment shall remain safe with performance characteristics as follows:

a. RF Power Out: 1 watt minimum with 50 percent modulation capability.

b. Receiver Sensitivity: 6 microvolts (open circuit) maximum for 10 dB signal plus noise to noise ratio.

c. Audio Output Power: Not greater than 6 dB below standard condition.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:37Z
Check the source to verify that this is the currentversion before use.

The emergency power source may impose an interval of zero voltage at time of loss of primary power for a maximum period of 30 seconds. The equipment is not required to perform during this period of power loss but shall return to performance as specified above after return to emergency power conditions in accordance with MIL-STD-704.

3.5.1.11 Electrical Overload Protection - The radio set shall be self protecting against overload (above reserve capability) due to electrical faults and overloads within the equipment. The radio set shall be provided with overload protection by means of a 5 ampere fuse conforming to the requirements of MIL-F-15160. The fuse shall be readily accessible. A spare fuse shall be provided in the radio set. Only one fuse shall be required and this fuse shall be used in the plus (+) DC input power line as close as possible to the power input connector.

3.5.1.12 Electrical Connections - Connections to external circuits shall be provided as follows and shown in Publication EI-603:

Reference Designation	Receptacle Type (or approval equivalent)	Function
J1	BNC	RF Antenna
J2	Bendix 21-900129-1	Main Connector

3.5.1.13 Warm-up Time - The time required for the radio set to warm-up prior to operation shall be kept to a minimum and shall be not greater than 1 second for reception and 5 seconds for transmission.

3.5.1.14 Frequency Accuracy - The transmitted RF carrier frequency shall be within ± 2.0 kHz of the selected operating frequency. The frequency of the main receiver shall be within ± 2.0 kHz of the selected operating frequency. The frequency of the guard receiver shall be within ± 5.0 kHz of the internally selected operating frequency.

3.5.1.15 Channel Changing -

3.5.1.15.1 Channel Changing Time - The time from the instant any channel is selected (after warm-up) to the time that the radio set is operative for either transmission or reception shall be not greater than 0.3 seconds.

3.5.1.15.2 Spontaneous Channel Change - When operating, there shall be no spontaneous change of channels when the equipment is subject to any combination of environmental conditions or electrical input in accordance with MIL-STD-704.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.1.16 Antenna Circuits - The main receiver, transmitter, and guard receiver shall all operate with the same antenna through connector J1. The receivers shall be so isolated during transmission that they are not damaged and no degradation in the performance except as specified herein shall result from using a common antenna. No damage shall occur to any part of the radio set when the transmitter is operated with the RF output shorted, open-circuited or terminated in any complex impedance between these two conditions.

3.5.1.17 Main and Guard Receiver Characteristics - Unless otherwise specified herein, the following receive characteristics shall apply to both the main and the guard receiver. All receiver RF input levels specified herein shall be open circuit. Open circuit is defined as the test signal generator output level setting which is then coupled to the receiver RF input through a 6 dB attenuator. Closed circuit is defined as the voltage level actually measured across the input without reference to the test signal generator output level setting.

3.5.1.17.1 Receiver RF Input Termination - The main and guard receivers shall meet the specified performance requirements with an RF input having a source impedance of 50 $\pm$ 7 ohms.

3.5.1.17.2 Input Signal Protection - The main and guard receivers shall not block with inputs up to 1.0 volt and shall not be damaged by inputs up to 10 volts (on channel).

3.5.1.17.3 Noise Peak Limiter - The detector of the main and guard receivers (following the intermediate frequency amplifier) shall be provided with an instantaneous noise peak limiter. The noise limiting action shall effectively limit the audio output voltage of modulating surges or peaks exceeding 50 $\pm$ 15 percent modulation of the RF signal. Its action shall be rapid enough to prevent loss of intelligibility due to reduction in receiver gain under the influence of continuous RF noises such as are caused by commutators, ignition systems, etc.

3.5.1.17.4 Internal Noise Level - The signal plus noise to noise (modulated to unmodulated) ratio of the receivers shall be at least 30 dB when a 1,000 microvolt (open circuit) signal modulated 30 percent at 1,000 Hz is applied and the audio output (signal plus noise) is adjusted to 250 milliwatts.

3.5.1.17.5 Intermediate Frequency Rejection - The main and guard receivers shall provide an attenuation of at least 80 dB to a signal on the intermediate frequencies relative to the response of the desired signal.

3.5.1.17.6 Regeneration - The interstage and overall regeneration of the main and guard receiver shall be so minimized that no evidence of instability shall occur at maximum gain.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.1.17.7 Receiver Amplitude Variation - The main and guard receiver audio output shall not vary more than +2, -3 dB for 30 percent modulated (at 1,000 Hz) RF inputs from 3.0 microvolts to 500 millivolts (open circuit) with the reference at 1,000 microvolts (open circuit). For levels above 500 millivolts see 3.5.1.17.2.

3.5.1.17.8 Squelch - The main and guard receivers shall be provided with a carrier to noise type squelch capable of operating, at the sensitivity levels specified in 3.5.1.17.9 with a signal plus noise to noise ratio adjustable between 5 dB and 10 dB. With the squelch circuit operating, the audio output of the main receiver shall be at least 40 dB down from the 250 milliwatt output level of 3.5.1.17.13.2. With the squelch set to be open on a 3 microvolt (open circuit) signal modulated 30 percent at 1 kHz, and with this signal applied at the receiver input, the attack time shall be 65 milliseconds maximum and the release time shall be 275 milliseconds maximum. The attack time shall be defined as the time interval between the application of the signal at the receiver input and full receiver audio output. The release time shall be defined as the time interval between the removal of the signal at the receiver input and fully squelched receiver audio output. The change in RF input level between attack and release points shall not exceed 4 dB.

3.5.1.17.8.1 Squelch Override - The main receiver and guard receiver squelch circuits shall contain an adjustable override function for recognition of CLIMAX system or other special condition signals, such that for all RF signal levels above the adjusted range of 20 ± 10 microvolts (open circuit) the squelch shall operate as an automatic gain control (AGC) carrier type squelch. The main and guard receiver squelch shall operate when receiving emergency beacon signals from the AN/URT-33, AN/PRC-90, or equivalent transmitters.

3.5.1.17.9 Sensitivity - With the squelch enabled, and the receiver audio output level adjusted to 250 milliwatts for an input of 1,000 microvolts (open circuit) modulated 30 percent at 1 kHz, the sensitivity of the main and guard receivers under standard conditions and over the specified frequency range shall be such that an RF input signal of not more than 3.5 microvolts (open circuit) modulated 30 percent at 1,000 Hz applied at the antenna terminals of the radio set shall produce a minimum (modulated to unmodulated) signal plus noise to noise ratio of 10 dB and a minimum audio sensitivity shall be measured with the signal generator, with an effective source impedance of 52 ohms, connected to the antenna terminal of the radio set. A signal level of not more than 5.0 microvolts (open circuit) shall produce a minimum signal plus noise to noise ratio of 10 dB under service conditions. When the main receiver is tuned to frequencies within +10 MHz of the guard frequency, or to 380 MHz or above, the main receiver shall meet the sensitivity requirements within 4.0 microvolts (open circuit) under standard conditions and 6 microvolts (open circuit) under service conditions. When the main receiver is tuned to frequencies within -10 MHz, +20 MHz of the guard frequency, the guard

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

receiver shall meet the sensitivity requirements with 4.0 microvolts (open circuit) under standard conditions and 6 microvolts under service conditions. Not more than 5 total main receiver channels out of 7,000 may be degraded from the above sensitivity. In the event a degraded channel exists, its frequency and sensitivity shall be reported to the procuring activity for inclusion in handbooks and support documentation.

3.5.1.17.10 Selectivity - The overall selectivity of the receiver shall be within the following limits:

Mode	6 dB Bandwidth	60 dB Bandwidth
AM Normal (Main and Guard Receiver)	38 kHz Minimum	80 kHz Maximum
AM Secure (Main and Receiver only)	70 kHz Minimum	170 kHz Maximum

Passband ripple shall not exceed 1.5 dB over 90 percent of the 6 dB bandwidths.

3.5.1.17.11 Cross Modulation and Desensitization - Cross modulation and desensitization effects of interfering signals (modulated 30 percent at 1,000 Hz) on channels more than one (1) MHz from and at a level 75 dB above the desired channel reference level (3.0 microvolts open circuit at 10 dB signal plus noise to noise) shall permit a signal plus noise to noise ratio of the desired signal of at least 6 dB.

3.5.1.17.12 Image and Spurious Rejection - Image rejection shall be at least 60 dB relative to the response to the desired signal. Spurious rejection shall be at least 70 dB in the narrowband mode and 60 dB in the wideband mode relative to the response of the desired signal.

3.5.1.17.13 Normal Audio Output - The main and guard receivers shall have a common audio output. This common output shall be the narrowband audio output of the main receiver with the operating output level adjustable by the control described in 3.5.1.5.4.

3.5.1.17.13.1 Normal Audio Output Impedances - The normal AM audio output impedances shall be such that the audio characteristics specified herein shall be met when loaded into 150 or 600 ohms respectively. This output shall be floating with respect to chassis ground. External grounding any one of the output leads shall have no adverse effects on receiver performance. Both 150 and 600 ohms output lines shall be available at separate pins on the external connector.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.1.17.13.2 Normal Audio Output Level - The receiver audio output shall be internally adjustable from less than 25 to at least 250 milliwatts with the control described in 3.5.1.5.4 in full clockwise position and an RF input of 1,000 microvolts (open circuit) modulated 30 percent at 1,000 Hz. Unless specified otherwise, the internal adjustment shall be set for 250 milliwatts, within +20 percent, -10 percent. Under service conditions, the audio level shall not vary more than 3 dB from the level set under standard conditions.

3.5.1.17.13.3 Normal Audio Output Response - The receiver audio output shall be within +1 and -3 dB of a 1,000 Hz reference for modulation signals between 300 Hz and 3,500 Hz. Below 300 Hz and above 3,500 Hz, the audio response shall decrease at 6 dB per octave minimum.

3.5.1.17.13.4 Normal Audio Output Distortion - Total harmonic distortion at the receiver normal audio output shall not exceed 10 percent for 30 percent modulation at 1,000 Hz with input signal levels from 100 microvolts to 250,000 microvolts (open circuit).

3.5.1.17.13.5 Normal Audio Output Phase - The audio output of the guard receiver shall be in phase with the main receiver audio output at the radio output terminals.

3.5.1.17.13.6 Unattenuated Normal Audio Output - In addition to the normal audio outputs loaded into 150 or 600 ohms with operator control of the output level, an unattenuated (not operator controlled) normal audio output shall be provided on a separate pin on the external connector. This output source impedance shall be less than 100 ohms. With the 150 ohm or 600 ohm output properly terminated, the unattenuated output level shall be 6.5 $\pm$ 1.0 volts RMS for an RF input of 1,000 microvolts (open circuit) modulated 30 percent at 1,000 Hz. Frequency response shall be within +1 and -3 dB of a 1,000 Hz reference for modulation signals between 300 Hz and 3,500 Hz. Distortion shall not exceed 10 percent at 1,000 Hz.

3.5.1.17.14 Main Receiver Characteristics - The main receiver shall meet all requirements of 3.5.1.17 through 3.5.1.17.13.6 plus the following when the function selector switch is in the MAIN, BOTH or ADF position. The requirements of 3.5.1.17.15 do not apply to the main receiver.

3.5.1.17.14.1 Wideband Audio Outputs - The main receiver shall have two wideband audio outputs in addition to the normal audio outputs. One wideband audio output shall be for use with TSEC/KY-28 secure voice systems. The other shall be for use with ADF systems. Receiver signals for these outputs shall not pass through the carrier to noise squelch circuit and shall be operative at all times when in the receive mode.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.1.17.14.1.1 Wideband Audio Output Impedances - The output impedance for the TSEC/KY-28 system shall be 1,000 ohms maximum at 1,000 Hz and shall provide proper output when operated into an unbalanced 500 ohm or higher impedance load. The output impedance for the ADF system shall be 20,000 ohms maximum at 1,000 Hz and shall provide proper output when operated into a 20,000 ohm ± 10 percent resistive load.

3.5.1.17.14.1.2 Wideband Audio Output Levels - The TSEC/KY-28 system output level shall be 0.250 volts RMS minimum across 500 ohms with an RF input of 1,000 microvolts (open circuit) modulated 30 percent at 1,000 Hz. The ADF system output level shall be 4.0 volts peak to peak minimum across 20,000 ohms with an RF input of 1,000 microvolts (open circuit) modulated 30 percent at 1,000 Hz.

3.5.1.17.14.1.3 Wideband Audio Output Responses - The receiver wideband audio output for TSEC/KY-28 shall be within +2 to -4 dB of the 1,000 Hz reference level for modulating signals between 300 Hz and 25 kHz. The frequency response of the ADF audio shall be within ± 2 dB of the 1,000 Hz reference level for modulating signals between 50 Hz and 10 kHz.

3.5.1.17.14.1.4 Wideband Audio Output Distortion - Total harmonic distortion at the receiver wideband audio outputs shall not exceed 10 percent for 50 percent modulation at 1,000 Hz with signal levels up to 250,000 microvolts (open circuits).

3.5.1.17.14.1.5 Wideband Audio Output Phase - The ADF audio output shall be not more than 20 degrees out of phase with the negative portion of the RF signal modulation envelope when the RF signal is modulated 30 percent at 200 Hz.

3.5.1.17.15 Guard Receiver Characteristics - The guard receiver shall meet all requirements of 3.5.1.17 through 3.5.1.17.13.6 plus the following when the function selector is in the BOTH or ADF positions. The requirements of 3.5.1.17.14 do not apply to the guard receiver. The guard receiver shall be capable of operation on a fixed frequency channel in the RF range 238.000 to 248.000 MHz changeable in 25 kHz increments. The normal center frequency shall be 243.000 MHz. It shall be possible to change the guard receiver center frequency by crystal replacement and manual retuning in the maintenance shop. The guard receiver shall be required to receive AM voice and AM pulse tone modulated RF signals. The guard receiver shall share the RF antenna input with the main receiver. Both the main and guard receiver shall be able to receive signals simultaneously when the antenna changeover switch is in the receive condition. In all other positions of the function selector switch, except the BOTH and ADF positions, the guard receiver shall be off and not require any line current.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the currentversion before use.

3.5.1.17.15.1 Low Level Guard Audio Output - A separate line shall be provided for a low level guard receiver output. The level shall be between 0.15 V and 0.45 V RMS into a 600 ohm ± 10 percent resistive load at a 3 microvolt (open circuit) input signal, 30 percent modulated at 1 kHz. This guard receiver output is for use with the TSEC/KY-28 and shall be terminated in a 600 ohm ± 10 percent resistive load. With a reference input of 1,000 microvolts 30 percent modulated at 1 kHz, the frequency response shall be within +1 to -3 dB from 300 to 3,500 Hz and distortion at 1,000 Hz shall not exceed 10 percent.

3.5.1.18 Transmitter Characteristics - The transmitter shall provide an amplitude modulated RF carrier frequency for voice (normal and secure) or tone modulation inputs.

3.5.1.18.1 Transmitter RF Termination - The transmitter shall meet the specified performance requirements with an RF termination of 50 ohms ± 7 ohms.

3.5.1.18.2 RF Power Output - At ambient temperatures below 55°C and under all service conditions except low input voltage, the unmodulated RF power output shall be not less than 10 watts. Under reduced input voltage conditions, the unmodulated power output may be reduced as follows:

27.5 volts to 26.0 volts, 10 watts - 0.5 dB
26.0 volts to 24.0 volts, 10 watts - 1.0 dB

In ambient temperatures above +55°C the unmodulated RF power output may decrease but shall be not less than 5 watts at +71°C. Under standard conditions when the transmitter is connected to any impedance producing a VSWR of 2.5 to 1 maximum, the RF power output shall be not less than 7 watts. The thermal protection circuit of 3.5.1.9 may reduce the RF power output to 1 watt minimum for any required combination of extreme service conditions, VSWR, and extended duty cycle. When the same frequency channel is selected several times in succession, the transmitter RF power output shall not vary more than 0.5 dB from the power obtained initially.

3.5.1.18.3 RF Carrier on Time - The RF carrier shall reach 90 percent of the required output level within 50 milliseconds after actuation of the external transmit (key) switch when the function selector switch is in either MAIN, ADF, or BOTH position.

3.5.1.18.4 Transmitter Audio Input - Separate audio inputs shall be provided for normal voice, secure voice, and retransmit signals to be transmitted. The transmitter modulator circuit shall include a tone generator for modulating the transmitter with a 1,020 Hz ± 10 percent frequency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.1.18.4.1 Audio Input Impedances - The normal voice audio input impedance shall be 150 ohms ± 10 percent unbalanced. The secure voice audio input impedance shall be 2,000 ohms ± 30 percent unbalanced at 1,000 Hz. The retransmit audio input impedance shall be 500 ohms minimum unbalanced.

3.5.1.18.4.2 Microphone Bias Voltage - The normal voice audio input shall provide bias voltage and current to operate a carbon microphone or amplifier (AM-3597B/A type) for a dynamic microphone. With normal input power to the radio set, the microphone bias current supplied shall be 55 ± 10 milliamperes when measured with an external 100 ohm resistive load on the normal voice audio input. The voltage supplied shall be 22 volts DC minimum with no load (100 ohm resistor removed) and the source impedance shall not exceed 450 ohms. Under emergency power or extreme service conditions, the no load voltage shall be at least 12 volts DC.

3.5.1.18.4.3 Transmitter Audio Levels - The normal voice audio input shall be designed for 0.500 volts RMS (closed circuit) input signals. The secure voice audio input shall be designed for 12 volt peak to peak (closed circuit) input signals. The retransmit audio input shall be designed for 6.5 volt RMS (closed circuit) input signals.

3.5.1.18.4.4 Transmitter Audio Frequency Range - The normal voice audio input shall be designed for input signals between 300 and 3,500 Hz. The secure voice audio input shall be designed for input signals between 300 and 25,000 Hz. The retransmit audio input shall be designed for input signals between 300 and 3,500 Hz.

3.5.1.18.5 Transmitter Modulation Characteristics - The audio modulation of the transmitter RF carrier output shall have the following characteristics:

3.5.1.18.5.1 Percent Modulation - Under standard conditions the percent modulation shall be between 90 and 100 percent for the 1,000 Hz tone, a 0.5 volt RMS (1,000 Hz) closed circuit signal at the normal voice audio input, a 6.5 volt RMS (1,000 Hz) closed circuit signal at the retransmit audio input, or a 12 volt peak to peak (1,000 Hz) closed circuit signal at the secure voice audio input. Under environmental conditions, the percent modulation shall not be less than 70 percent. The percent modulation shall be determined from an oscilloscope presentation of the modulated RF carrier and measurement of maximum (Emax) and minimum (Emin) voltages displayed on the oscilloscope. Modulation percentages shall then be calculated from the following formula:

$$\text{Percent Modulation} = \frac{\text{Emax} - \text{Emin}}{\text{Emax} + \text{Emin}} \times 100 \text{ percent}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.1.18.5.2 Carrier Fidelity - With a constant normal voice audio input voltage of 0.5 volts RMS closed circuit adjusted to any frequency between 300 and 3,500 Hz, the demodulated audio carrier voltage shall be within +1 dB, -3 dB of the demodulated audio carrier voltage resulting from a 0.5 volt closed circuit 1,000 Hz audio input. With a constant retransmit audio input voltage of 6.5 volts RMS closed circuit adjusted to any frequency between 300 and 3,500 Hz, the demodulated audio carrier voltage shall be within +1 dB, -3 dB of the demodulated audio carrier voltage resulting from a 6.5 volt closed circuit 1,000 Hz audio input. With a constant secure voice audio input voltage of 12 ±1 volts peak to peak (closed circuit) adjusted to any frequency between 300 and 25,000 Hz, the demodulated audio carrier voltage shall be within ±3 dB of the demodulated output at 1,000 Hz modulation from 300 Hz to 1 kHz, within ±2 dB from 1 kHz to 20 kHz and shall be attenuated at 6 dB per octave minimum above 25 kHz.

3.5.1.18.5.3 Carrier Distortion - With a normal voice audio input of 0.35 volts RMS closed circuit adjusted to any frequency between 300 and 3,500 Hz, a retransmit audio input of 4.5 volts RMS closed circuit adjusted to any frequency between 300 and 3,500 Hz or a secure voice audio input of 3.0 volts RMS closed circuit adjusted to any frequency between 300 and 25,000 Hz, the demodulated audio harmonic distortion shall not exceed 10 percent.

3.5.1.18.5.4 Carrier Noise Modulation - The audio voltage detected from an unmodulated carrier shall be at least 40 dB below the detected audio voltage obtained from a carrier modulated 90 percent at 1,000 Hz.

3.5.1.18.5.5 Carrier Frequency Modulation - With the RF carrier amplitude modulated 90 percent at 1,000 Hz, the peak frequency deviation arising from incidental frequency modulation (FM) of the transmitter shall not exceed ±2.0 kHz.

3.5.1.18.6 Transmitter Sidetone - A transmitter sidetone shall be developed by sampling a portion of the modulated transmitter carrier in the antenna circuit. The sidetone signal shall be introduced into the audio portion of the main receiver and adjustable from 25 to 250 milliwatts.

3.5.1.18.6.1 Sidetone Level - Sidetone level shall be so internally adjustable that with a normal voice audio input voltage of 0.5 volts closed circuit at 1,000 Hz, the sidetone power output at the output of the main receiver shall be 250 milliwatts, +0 dB to -2 dB.

3.5.1.18.6.2 Sidetone Frequency Response - With a constant normal audio input voltage of 0.5 volts closed circuit adjusted to any frequency between 300 and 3,500 Hz, the sidetone audio output voltage of the radio set shall be within +1 dB, -5 dB, of the sidetone output voltage at the 1,000 Hz audio input.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.1.18.6.3 Sidetone Distortion - Using any normal audio input frequency between 300 and 3,500 Hz and with the audio input adjusted for 40 percent modulation, the sidetone distortion shall not exceed 15 percent.

3.5.1.19 Automatic Relaying Operation - With the function selector switch in MAIN or BOTH position, the main receiver and transmitter of the radio set shall operate in an automatic relaying mode (both normal and secure voice) when used in conjunction with another AN/ARC-159 or AN/ARC-159(V) radio set plus the C-9451/ARC-159(V) and SA-1964/ARC-159(V) or C-9960/ARC-159(V) and SA-2078/ARC-159(V). No special switch position shall be provided on the radio set front panel to indicate the relay mode of operation. The radio set shall provide at pins of the main power connector the following inputs, outputs, and controls that allow the radio set to be used for automatic relaying when the function selector switch is in the MAIN or BOTH positions:

a. Main receiver audio outputs
b. Retransmit audio inputs
c. Automatic relay trigger output
d. Transmit control (T/R control line)

The application to the antenna circuit of either radio set, of a signal on the receiver selected frequency and of a level sufficient to open the receiver squelch circuit shall cause the other radio set to automatically generate and emit a radio frequency carrier wave of the units rated power on its selected frequency. Removal of the actuating receiver signal shall cause the cessation of the generated carrier. When the actuating signal is 3 dB or more above the level required for relaying as described above, the generated carrier shall reach 90 percent of rated power level within 125 milliseconds after application of the actuating signal and the system shall be returned to standby conditions within 300 milliseconds after removal of the actuating signal. The radio sets shall perform the relay function on any two RF channels within the tuning range provided a frequency separation of at least 10 MHz exists between the two channels with an isolation of at least 60 dB between the two antennas.

3.5.1.20 ADF Operation - With the function selector switch in the ADF position, the radio set shall operate in conjunction with a Radio Direction Finding Group to provide bearing information. When used for this service the receiver shall amplify the signals received on the ADF antenna, demodulate them and deliver them to the ADF group. ADF antenna protection from transmitter RF power is achieved by relays external to (not part of) this radio set in conjunction with the key line delay specified in 3.5.1.18.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.1.21 TSEC/KY-28 Operation - With the function selector switch in MAIN or BOTH position the radio set shall operate in conjunction with the TSEC/KY-28 speech security equipment described in CSEEB-13 and KAM-213/TSEC. Any specific circuitry of the radio set requiring activation for TSEC/KY-28 secure voice operations shall be actuated by applying an electrical ground to two separate pins (as appropriate) on the radio set connector. Grounding one of these pins shall switch the receiver IF to the wide bandwidth. Grounding the other provides volume control of audio from an external source using the radio set volume control of 3.5.1.5.4.

3.5.2 Radio Set AN/ARC-159(V)1 - The radio set shall meet all requirements of 3.5.1 plus the following:

3.5.2.1 Function - The radio set shall provide all the functions and capabilities of 3.5.1 plus the following capabilities:

a. Transmission of signals from a reference signal generator of CASS or DICASS on an assigned channel.

b. Compatibility with remote displays of 3.5.7.

c. Compatibility with the remote control of 3.5.4.

d. Guard channel precedence operation whereby application of an electrical ground to the assigned pin on the external connector automatically turns the radio set on and tunes it to the guard channel.

NOTE: Either one remote control unit or one remote frequency-channel indicator may be operated from the primary power supplied through the radio set. For any additional loading the primary power must be supplied directly from the aircraft power by relay or equivalent radio controlled switching.

3.5.2.2 Form Factor - The radio set shall not exceed 9.25 inches by 5.75 inches by 4.875 inches (including control knobs and integral connectors) as shown in EI-603.

3.5.2.3 Weight - The weight of the radio set shall be not greater than 9.25 pounds.

3.5.2.4 Contents - The radio set shall be one self contained unit of all solid state modular electronic circuit design containing the same modules and assemblies requirements of 3.5.1, but including additional circuitry required for capabilities listed in 3.5.2.1.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.2.5 Controls - Controls required for operation of the radio set shall be located on the Control/Indicator Assembly at the front of the radio set, and be functionally equivalent to those listed in 3.5.1.5. The "READ" switch shall initiate presentation of the preset channel frequency. A READ position is not required on the GUARD-MANUAL-PRESET switch in the AN/ARC-159(V)1, if this function is available as a separate switch. Additionally, the radio set shall have provisions for remote control by the control described in 3.5.4. For remote control applications, the frequency and channel selection information shall be in serial digital format such that the information transfer can be accommodated by two pairs of wires. One pair shall handle the clock reference, pluses and the other pair shall convey the serial digital information. The clock rate shall be approximately 800 Hz, and the information contained in a 32 bit message continually repeated. For remote control operations, a separate input shall be provided such that application of an electrical ground shall disable the controls on the front panel and allow total control by the equipment described in 3.5.4. The front panel control functions shall not affect radio set performance when being remotely controlled.

3.5.2.6 Preset Channel Memory Device - A memory device providing the capability described in 3.5.1.6 shall be included in the radio set. This memory device may have either mechanical memory storage or electronic memory storage.

3.5.2.7 Frequency Chart - A chart identical to that described in 3.5.1.7 shall be provided in the radio set.

3.5.2.8 Frequency/Channel Display - A display identical to that described in 3.5.1.8 shall be provided in the radio set. Additionally, the radio set shall provide serial data output to operate equivalent displays of a remote control unit as described in 3.5.4 and/or remote indicators as described in 3.5.7. The clock and data pulses shall be as defined in 3.5.2.5 and primary power accommodations for the remote units shall be provided as noted in 3.5.2.1. When operated from a remote control unit, the AN/ARC-159(V)1 is not required to display frequency or channel information duplicating that of the remote control unit.

3.5.2.9 Duty Cycle - The requirements of 3.5.1.9 shall apply.

3.5.2.10 Emergency Power - The requirements of 3.5.1.10 shall apply. During this condition, power to operate a remote control and indicators shall be maintained.

3.5.2.11 Electrical Overload Protection - The requirements of 3.5.1.11 shall apply. A 5 ampere fuse and spare shall be provided in the radio set.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.2.12 Electrical Connections - Connections to external circuits shall be provided on receptacles as follows and shown in EI-603.

Reference Designation	Receptacle Type (or approved equiv.)	Function
J1	BNC	RF Antenna
J2	Bendix 21-900129-1	Main/Power
J3	Bendix 21-900129-2	Remote/Power

3.5.2.13 Warm-Up Time - The requirements of 3.5.1.13 shall apply.

3.5.2.14 Frequency Accuracy - The requirements of 3.5.1.14 shall apply for either internal or remote control applications.

3.5.2.15 Channel Changing - The requirements of 3.5.1.15 shall apply for either internal or remote control applications.

3.5.2.16 Antenna Circuits - The requirements of 3.5.1.16 shall apply.

3.5.2.17 Main and Guard Receiver Characteristics - The requirements of 3.5.1.17 shall apply. Additionally, the equipment shall be turned on and the main receiver shall automatically tune to the guard channel upon application of an external electrical ground to the assigned pin.

3.5.2.18 Transmitter Characteristics - The requirements of 3.5.1.18 shall apply. Additionally, provisions shall be made for transmission of signals from a reference signal generator of CASS or DICASS, and automatic tuning of the transmitter to the guard channel upon application of an external electrical ground to the assigned pin.

3.5.2.18.1 CASS/DICASS Input Characteristics - A separate audio input shall be provided for CASS or DICASS signals to be transmitted. The CASS/DICASS interface will provide the external transmit (key) switch actuation at least 50 milliseconds prior to input of CASS/DICASS signals to the radio set.

3.5.2.18.1.1 CASS/DICASS Input Impedance - The input impedance presented by the radio set to the CASS/DICASS signal generator shall be 100 ohms +5 percent resistive.

3.5.2.18.1.2 CASS/DICASS Signal Levels - The CASS/DICASS input shall be designed for 0.5 volt RMS (closed circuit) input signals.

3.5.2.18.1.3 CASS/DICASS Input Frequency Range - The CASS/DICASS input shall be designed for input signals between 1 kHz and 50 kHz.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.2.18.2 CASS/DICASS Modulation Characteristics - CASS/DICASS signal modulation of the transmitter RF carrier output shall have the following characteristics.

3.5.2.18.2.1 Percent Modulation - Under standard conditions the percent modulation shall be between 85 and 100 percent for a 0.5 volt RMS (1 kHz to 50 kHz) closed circuit signal at the CASS/DICASS input. Under environmental conditions, the percent modulation shall not be less than 70 percent.

3.5.2.18.2.2 Carrier Fidelity - With a constant CASS/DICASS input voltage of 0.5 volts RMS closed circuit adjusted to any frequency between 1 kHz and 50 kHz, the demodulated signal voltage shall be constant within +2 dB.

3.5.2.18.2.3 Modulation Distortion - With a CASS/DICASS input of 0.35 volts RMS closed circuit adjusted to any frequency between 1 kHz and 50 kHz, the demodulated signal total harmonic distortion shall not exceed 10 percent.

3.5.2.18.2.4 Intermodulation Distortion - The demodulated level of the sum and difference frequencies of two equal level .35 volts RMS (open circuit) discrete (non-harmonically related) modulating CASS/DICASS signals shall be 25 dB below either modulating signal.

3.5.2.18.2.5 Carrier Noise Modulation - The requirements of 3.5.1 18.5.4 shall apply.

3.5.2.18.2.6 Carrier Frequency Modulation - The requirements of 3.5.1.18.5.5 shall apply.

3.5.2.18.2.7 Phase Linearity - The phase response of the modulating frequency shall be linear with frequency within +15 degrees throughout the 1 kHz to 50 kHz spectrum. In addition, the phase versus frequency slope shall not exceed 3.5 degrees per Hz.

3.5.2.19 Automatic Relaying Operation - The requirements of 3.5.1.19 shall apply.

3.5.2.20 ADF Operation - The requirements of 3.5.1.20 shall apply.

3.5.2.21 TSEC/KY-28 Operation - The requirements of 3.5.1.21 shall apply.

3.5.2.22 CASS/DICASS Operation - With the function selector switch in MAIN or BOTH position, the radio set shall operate in conjunction with the CASS/DICASS reference signal generator described in MIL-G-81665 to transmit signals provided by the signal generator. The CASS/DICASS interface will provided the external transmit (key) swith operation at least 50 milliseconds prior to input of CASS/DICASS signals to the radio set.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the currentversion before use.

3.5.2.23 Guard Precedence Operation - The radio set, under normal operating conditions shall have front panel control facilities as described in 3.5.2.5. This radio set shall also have provisions for guard precedence frequency selection from an external switch. This operating mode, intended for emergency condition use, shall become effective by the closing of an external switch (not included as part of the radio set) to provide an electrical ground on the assigned pin. When this switch is closed (via one wire to the radio set) the front panel controls shall be disabled, the radio set turned on and the main transceiver tuned to the guard frequency. When this operation is effective, the control panel and remote indicator, if used, shall indicate the guard frequency.

3.5.3 Receiver-Transmitters - For AN/ARC-159(V) system installations where separate control unit and receiver-transmitters are used, either the RT-1150/ARC-159(V) or the RT-1194/ARC-159A(V) shall provide the specified receiver-transmitter functions. The RT-1150/ARC-159(V) shall have a transmitter RF carrier output capability of ten watts and the RT-1194/ARC-159A(V) shall have a transmitter RF output capability of thirty watts. Details specifications for the RT-1150/ARC-159(V) and RT-1194/ARC-159A(V) are covered in paragraphs 3.5.3.1 and 3.5.3.2 respectively.

3.5.3.1 Receiver-Transmitter, Radio RT-1150/ARC-159(V) - The receiver-transmitter shall meet all the performance requirements of 3.5.1 as modified by 3.5.2, except for those paragraphs referring to control functions and displays being located in the same unit.

3.5.3.1.1 Function - The receiver-transmitter shall provide all the functions and capabilities of 3.5.1.1 as modified by 3.5.2.1 except that control of the modes or functions and display of the radio frequency or preset channel number shall not be provided within, but rather remotely controlled and displayed by the units in 3.5.4.

3.5.3.1.2 Form Factor - The receiver-transmitter shall be not greater than 9.0 inches by 5.0 inches by 4.875 inches (including integral connectors and handle) as shown in EI-603.

3.5.3.1.3 Weight - The weight of the receiver-transmitter shall be not greater than 8.0 pounds.

3.5.3.1.4 Contents - The receiver-transmitter shall be one self-contained unit suitable for mounting outside the cockpit area, of all solid state modular electronic circuit design, having the same module and assembly requirements of 3.5.1 as modified in 3.5.2, but deleting the control/indicator assemblies for control of functions and modes or display of frequency and channel and including instead an additional adapter module required for compatibility with units described in 3.5.4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

and 3.5.7. The adapter module shall form the front surface of the receiver-transmitter and include provisions for securing it in any of the mounts described in 3.5.12 through 3.5.16. It shall also contain external plug in jacks for a headset and microphone for maintenance purposes. A volume control shall be provided for adjustment of the audio level at the included headset jack. The adapter module shall also contain accessible controls for squelch adjustment of the main and guard receivers.

3.5.3.1.5 Controls - Controls for normal operation of the receiver-transmitter shall be located on the control described in 3.5.4. Display of the operating radio frequency and preset channel number shall be on the remote control or indicator described in 3.5.7. The input and output frequency and channel information shall be identical to that required in 3.5.2.5. For installations using two remote controls for one receiver-transmitter, an external switch is required to designate which control has effectivity.

3.5.3.1.6 Preset Channel Memory Device - A memory device is not required in the receiver-transmitter, but rather on the control described in 3.5.4.

3.5.3.1.7 Frequency Chart - A frequency chart is not required on the receiver-transmitter.

3.5.3.1.8 Frequency/Channel Display - A frequency/channel display is not required on the receiver-transmitter.

3.5.3.1.9 Duty Cycle - The requirements of 3.5.1.9 shall apply, except that the receiver-transmitter shall operate satisfactorily under any of the environmental conditions or reasonable combination thereof as specified in MIL-E-5400 for Class 2 Curve A equipment when supplied with cooling air directed at the rear surface as shown in EI-603 for the following:

a. Duty Cycle, 1 minute transmit, 5 minutes receive, continuous operation

Cooling Air

Air Temperature °C	Flow Rate (pounds/hour)
20 degrees	10 pounds
40 degrees	15 pounds
60 degrees	45 pounds
71 degrees	100 pounds

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

b. Other operational/environmental capability of the RT-1150/ARC-159(V)

1. Duty cycles, 1 minute transmit, 2 minutes receive continuous operation. No cooling air shall be required up to +35°C temperature and altitude up to 10,000 feet, or up to 25°C temperature and altitude up to 30,000 feet.

2. Duty cycles, continuous transmit in a +71°C ambient temperature and altitudes up to 10,000 feet, or in a +35°C ambient temperature and altitude up to 40,000 feet.

Cooling Air Required

Air Temperature °C	Flow Rate (pounds/hour)
25 degrees	30 pounds
40 degrees	50 pounds

3.5.3.1.10 Emergency Power - The requirements of 3.5.1.10 shall apply. During these conditions, power to operate a remote control and indicators shall be maintained.

3.5.3.1.11 Electrical Overload Protection - The requirements of 3.5.1.11 shall apply. A 5 ampere fuse and spare shall be located in the receiver-transmitter.

3.5.3.1.12 Electrical Connections - Connections to external circuits shall be provided on receptacles as follows and shown in EI-603.

Reference Designation	Receptacle Type (or approved equiv.)	Function
J1	BNC	RF Antenna
J2	Bendix 21-900129-1	Main/Power
J3	Bendix 21-900129-2	Remote/Power
Mic Jack	JJ-033	
Headset Jack	JJ-034	

3.5.3.1.13 Warm-Up Time - The requirements of 3.5.1.13 shall apply.

3.5.3.1.14 Frequency Accuracy - The requirements of 3.5.1.14 shall apply when used in connection with the control described in 3.5.4.

3.5.3.1.15 Channel Changing - The requirements of 3.5.1.15 shall apply when using the control described in 3.5.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.3.1.16 Antenna Circuits - The requirements of 3.5.1.16 shall apply.

3.5.3.1.17 Main and Guard Receiver Characteristics - The requirements of 3.5.1.17 as modified by 3.5.2.17 shall apply. All references to controls or function selector switch positions (except squelch adjust) shall be as set on a control described in 3.5.4.

3.5.3.1.18 Transmitter Characteristics - The requirements of 3.5.1.18 as modified by 3.5.2.18 shall apply.

3.5.3.1.19 Automatic Relaying Operation - The requirements of 3.5.1.19 shall apply. All references to controls or function selector switch positions (except squelch adjust) shall be as set on a control described in 3.5.4.

3.5.3.1.20 ADF Operation - The requirements of 3.5.1.20 shall apply. All references to controls or function selector switch positions (except squelch adjust) shall be as set on the control described in 3.5.4.

3.5.3.1.21 TSEC/KY-28 Operation - The requirements of 3.5.1.21 shall apply.

3.5.3.1.22 CASS/DICASS Operation - The requirements of 3.5.2.22 shall apply.

3.5.3.1.23 Guard Precedence Operation - The requirements of 3.5.2.23 shall apply.

3.5.3.2 Receiver-Transmitter, Radio RT-1194/ARC-159A(V) - The receiver-transmitter shall meet the performance requirements of 3.5.3.1 except the transmitter RF power output requirements are increased to 30 watts RF output and for any other associated differences defined in this section.

3.5.3.2.1 Function - The receiver-transmitter shall provide all the functions and capabilities of the RT-1150/ARC-159(V) defined by 3.5.3.1.1. It shall operate with any control unit used to control the RT-1150/ARC-159(V).

3.5.3.2.2 Form Factor - The receiver-Transmitter shall not exceed 5 inches high by 6 inches wide by 14.3 inches long (including integral connectors and handles). The external power connectors and antenna connector shall be located on the front surface or receiver-transmitter. The receiver-transmitter shall have provisions for cooling by externally supplied air via the mounting defined in 3.5.17 or 3.5.18. Cooling versus environment is defined in 3.5.3.2.5 or 3.5.10. Configuration of the receiver-transmitter shall be as shown in EI-603.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the currentversion before use.

3.5.3.2.3 Weight - The weight of the receiver-transmitter shall be not greater than 13.5 pounds.

3.5.3.2.4 Contents - The receiver-transmitter shall be one self-contained unit suitable for mounting outside the cockpit area, shall be of solid state modular electronic circuit design and shall have the following module and assembly content.

Four modules shall be identical to those used in the RT-1150/ARC-159(V). These modules are:

a. Synthesizer Module
b. RF Translator/Modulator Module
c. IF/Audio Amplifier Module
d. Guard Receiver Module

The remaining portions of the receiver-transmitter shall consist of the following:

e. Power Amplifier/Power Supply Module - This module shall provide an equivalent function as the power amplifier/power supply module of the RT-1150/ARC-159(V), however, it shall have increased capability to provide the specified higher RF power output. This module shall have accommodations for cooling from externally supplied cooling air per 3.5.3.2.5 or the blower defined in 3.5.10.

f. Front Panel Assembly - The front panel assembly shall contain the main receiver-transmitter and guard receiver RF connectors and two power connectors as defined in 3.5.3.2.8.

   The front panel assembly shall contain an easily replaceable 12 ampere primary power fuse and a spare fuse. Additionally, controls and a headset jack shall be available for main and guard receiver squelch adjustment.

   An elapsed time meter with a 9999 hour range per MIL-M-7793 shall be located on the front panel with readout visible from the front of the unit.

g. Chassis and Interconnect Wiring Unit - The chassis shall provide the receiver-transmitter structural assembly for modules and shall contain the interconnect wiring provisions. Electrical components mounted to the chassis shall be held to a minimum and interconnect wiring shall be primarily through printed circuit board connections. The chassis and power amplifier/power supply module shall have provisions to operate from externally supplied cooling air.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.3.2.5 Cooling Air - The RT-1194/ARC-159A(V) shall operate satisfactorily under any combination of environmental conditions defined by MIL-E-5400 for Class 2, Curve A equipments when supplied with cooling air as defined below:

Air Temperature (°C)	Flow Rate (pounds/hour)
-15	27.2
- 5	27.5
5	29.0
15	32.3
25	37.8
35	46.4
45	58.0
55	85.0
71	250.0

a. Other environmental/operational capability of the RT-1194/ARC-159A(V)1 shall be as follows:

Continuous transmit operation, +25°C, altitude up to 30,000 feet.

Cooling Air

Temperature °C	Flow Rate (pounds/hour)
20 degrees	70 pounds
35 degrees	95 pounds
45 degrees	120 pounds

b. The receiver-transmitter shall be compatible with an air supply system having the following air flow-pressure characteristics.

Flow Rate (pounds/hour)	Pressure drop, inches of water (Reference density 0.0765 lb/cu ft.)
24	0.051
40	0.099
60	0.170
80	0.250
100	0.344

3.5.3.2.6 Duty Cycle - The requirements of 3.5.1.9 shall apply except that the duty cycle shall be 5 minutes maximum transmit and 10 minutes receive when supplied with cooling air per 3.5.3.2.5. The RF output under any required combination of extreme service conditions, voltage standing wave ratio (VSWR) and extended time transmission shall be 6 watts minimum and 50 percent minimum modulation capability.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:37Z
Check the source to verify that this is the currentversion before use.

3.5.3.2.7 Emergency Power - The requirements of 3.5.3.1.10 shall apply except that the transmitter RF power output shall be 6 watts minimum with 50 percent minimum modulation capability.

3.5.3.2.8 Electrical Connections - Connections to external circuits shall be provided on receptacles as follows and as shown in EI-603.

Reference Designation	Receptacle Type (or approved equiv.)	Function
J1	Type N	RF Antenna
J2	Bendix 21-3492-18-32P	Main/Power
J3	Bendix 21-3492-16-26S	Remote/Power
J4	Type TNC	Guard Receiver Antenna
J8	JJ-034	Headset Jack

3.5.3.2.9 Warm-Up Time - The requirements of 3.5.3.1.13 shall apply.

3.5.3.2.10 Frequency Accuracy - The requirements of 3.5.3.1.14 shall apply.

3.5.3.2.11 Channel Changing - The requirements of 3.5.3.1.15 shall apply.

3.5.3.2.12 Antenna Circuits - The main receiver-transmitter and guard receiver shall operate either from the common antenna connector, J1, or with the guard receiver connected independently to antenna connector, J4. The separate guard antenna input J4 shall be enabled when a ground is placed on the "guard receiver input select" pin of J3. The receivers shall be so isolated during transmission that they are not damaged. No degradation of performance, except as herein specified, shall result from the use of the common antenna. No damage shall occur in any part of the receiver-transmitter when the transmitter is operated with the antenna connection short circuited, open circuited or terminated in any complex impedance between these two extremes.

3.5.3.2.13 Main and Guard Receiver Characteristics - The requirements of 3.5.3.1.17 shall apply except for receiver sensitivity. The RT-1194/ARC-159A(V) sensitivity recognizing the alternate connections of J1 usage for both the main and guard receivers of J1 for the main receiver only and J4 for the guard receiver shall be as follows:

a. Operation with J1 connected to both the main and guard receivers:

   The sensitivity of the main receiver under standard conditions shall be such that an RF input of 3.5 microvolts (open circuit) shall produce a signal plus noise to noise ratio of at least 10 dB. The guard receiver shall meet this requirement with 4

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the current version before use.

microvolts input. When the main receiver is tuned within ±10 MHz of the guard receiver frequency the input level for main receiver sensitivity shall be 4.5 microvolts, and when tuned to 380 MHz or higher, the sensitivity shall be 4.0 microvolts. When the main receiver is tuned within -10 MHz to +20 MHz of the guard receiver frequency, the input level for guard receiver sensitivity shall be 5.5 microvolts.

Under service conditions the input levels for sensitivity measurements shall be:

Main Receiver: 5.0 microvolts except for frequencies within ±10 MHz of the guard receiver frequency the input shall be 5.5 microvolts.

Guard Receiver: 5.0 microvolts except when the main receiver is tuned within -10 MHz to +20 MHz of the guard channel frequency, the input shall be 6.5 microvolts.

b. Operation with J1 connected to the main receiver and J4 connected to the guard receiver-internally connected by grounding the antenna input select line.

The sensitivity of the main receiver, via J1 connector, under standard conditions shall be such that 3.5 microvolts (open circuit) shall produce a signal plus noise to noise ratio of at least 10 dB. When the main receiver is tuned to 380 MHz or higher the sensitivity shall be 4.0 microvolts. The sensitivity of the guard receiver via J4 connector under standard conditions shall be such that 3.5 microvolts (open circuit) shall produce a signal plus noise to noise ratio of at least 10 dB. Under service conditions the input for the respective main and guard receiver sensitivity requirements shall be 5 microvolts (open circuit).

3.5.3.2.14 Transmitter Characteristics - The requirements of 3.5.3.1.18 shall apply except as noted below:

3.5.3.2.14.1 RF Power Output - Under the conditions (standard and service) specified in 3.5.3.2.6, and with input voltage of 24 volts DC or higher, the unmodulated RF power output shall be 30 watts minus 0.4 dB minimum. Below 24 volts input the RF power may be automatically switched to a lower level, however, below 22 volts input, including emergency power conditions, the RF output shall be 6 watts minimum. The transmitter shall have automatic control of the output level to protect the power amplifier with mismatched load conditions. This control circuit shall start reducing power output with a VSWR of 1.5 to 1, nominal and continue reducing the power output with VSWR increase to a VSWR of 3.0 to 1, nominal. The rate of change of power output

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the currentversion before use.

shall not exceed the linear values of rated output at a VSWR of 1 to a -1.5 dB level at a VSWR of 2.5 to 1. At VSWR conditions beyond trip out (nominally 3 to 1) the power output shall be 6 watts minimum. When the same frequency channel is selected several times in succession, the transmitter RF power output shall not vary more than 0.5 dB from the power output obtained initially.

3.5.3.2.14.2 RF Carrier on Time - The requirements of paragraph 3.5.1.18.3 shall apply.

3.5.3.2.15 Automatic Relaying Operation - The requirements of 3.5.3.1.19 shall apply.

3.5.3.2.16 ADF Operation - The requirements of 3.5.3.1.20 shall apply.

3.5.3.2.17 TSEC/KY-28 Operation - The requirements of 3.5.3.1.21 shall apply.

3.5.3.2.18 CASS/DICASS Operation - The requirements of 3.5.3.1.22 shall apply.

3.5.3.2.19 Guard Precedence Operation - The requirements of 3.5.3.1.23 shall apply.

3.5.4 Controls, Radio Set - The control units of this section shall each provide complete control of the RT-1150/ARC-159(V) or the RT-1194/ARC-159A(V). Each of these controls may be also used as second or alternative control unit for the AN/ARC-159(V)1, the RT-1150/ARC-159(V) or the RT-1194/ARC-159A(V) with effectivity determined by an external "take-command" switch.

3.5.4.1 Controls, Radio Set C-9577/ARC-159(V), C-9577A/ARC-159(V), C-10436/ARC-159(V), and C-10436A/ARC-159(V) - The control shall meet the applicable requirements of MIL-C-6781 for Type I control panel. The lighting of the plastic panel shall conform to MIL-P-7788, Class 1-R (Navy Red) of 1-W (Air Force White in accordance with MIL-L-27160), Type V with a lamp voltage input of either 5 or 28 volts (AC or DC). Separate input power pins shall be provided for the 5 and 28 volt inputs such that application of power to the assigned pin provides proper illumination. Alternate panels are required for red or white lighting for the C-9577/ARC-159(V) and C-9577A/ARC-159(V). The lighting color shall be specified by the procurement document. The C-10436/ARC-159(V) and C-10436A/ARC-159(V) shall be IPL white lighted per MIL-C-25050 instead of MIL-P-7788 or MIL-L-27160.

3.5.4.1.1 Function - The control shall provide remote control of an equipment as described in 3.5.2 or 3.5.3 and be compatible with equipments described in 3.5.7.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:37Z
Check the source to verify that this is the currentversion before use.

3.5.4.1.2 Form Factor - The control shall be not greater than 5.25 inches by 5.750 inches by 4.875 inches (including knobs and integral connectors) as shown in EI-603.

3.5.4.1.3 Weight - The weight of the control shall be not greater than 3.5 pounds.

3.5.4.1.4 Contents - The control shall be one self-contained unit suitable for cockpit mounting with a Control/Indicator Asssembly identical to that provided for the equipment described in 3.5.2. The C-9577/ARC-159(V) and C-10436/ARC-159(V) shall compare to the AN/ARC-159(V)1, which has mechanical memory preset channel storage. The C-9577A/ARC-159(V) and C-10436A/ARC-159(V) shall compare to the AN/ARC-159(V)1, which has electronic memory preset channel storage. It shall also contain a Power Supply/Adapter Assembly that operates from +28 volts DC input power and provides interface of the Control/Indicator Assembly to the main external connector. The squelch adjust controls on the Control/Indicator Assembly shall not be operative.

3.5.4.1.5 Controls - Operating controls required to operate the radio set and receiver-transmitter shall be located on the Control/Indicator Assembly at the front of the control unit. The operating controls shall be functionally equivalent to those described in 3.5.1.5, except that 3.5.1.5.10 shall not be operative. The Power Supply/Adapter Assembly shall provide input and output frequency and channel information in a serial digital format such that information transfer between the associated radio set, receiver-transmitter, and indicators can be accommodated by two pairs of wires. All other control functions shall be accommodated by conventional wiring for the respective functions.

3.5.4.1.6 Preset Channel Memory Device - A memory device identical to that described in 3.5.2.6 shall be provided in the control unit.

3.5.4.1.7 Frequency Chart - A chart identical to that described in 3.5.1.7 shall be provided on the control unit.

3.5.4.1.8 Frequency/Channel Display - A display identical to that described in 3.5.1.8 shall be provided in the control unit.

3.5.4.1.9 Duty Cycle - The control unit shall be capable of continuous duty.

3.5.4.1.10 Emergency Power - The control unit input power (+28 volts DC) shall be supplied by the associated radio set or receiver-transmitter. The control unit shall operate with emergency input power supplied under the conditions of 3.5.1.10, 3.5.2.10, and 3.5.3.10. The control is not required to perform during periods of power loss, but shall return to performance as specified herein after return to emergency power conditions.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the currentversion before use.

3.5.4.1.11 Electrical Overload Protection - The control unit shall be electrically protected for overload conditions by the associated radio set or receiver-transmitter. A fuse is not required in the control unit.

3.5.4.1.12 Electrical Connections - Connections to external circuits shall be provided on a receptacle as follows and shown in EI-603.

Reference Designation	Receptacle Type (or approved equivalent)	Function
J1	MS27497E-18B-32P	Main/Power

3.5.4.1.13 Warm-Up Time - The time required for the control unit to warm-up prior to operation shall be not greater than 1.0 second and shall result in radio set or receiver-transmitter operation as specified in 3.5.1.13.

3.5.4.1.14 Frequency Accuracy - The requirements of 3.5.2.14 or 3.5.3.14 are applicable relative to the frequency selected.

3.5.4.1.15 Channel Changing - The requirements of 3.5.2.15 or 3.5.3.15 are applicable for channel changing from a remote control.

3.5.4.1.16 Remote Control Operations - The control unit shall provide remote control of an equipment as described in 3.5.2 and 3.5.3, and permit all functions, capabilities and operations specified there-under. In remote control operations, all references to controls or function selector switch positions (except squelch adjust) shall be as set on the remote control. For remote control of the radio set, an external switch is required to select which unit will control operation. For dual remote control of the receiver-transmitter, an external switch is required to select which unit will control operation.

3.5.4.1.17 Remote Frequency/Channel Display Operations - The C-9577/ARC-159(V) and C-10436/ARC-159(V) series control unit shall display the operating frequency or preset channel number derived from signals supplied by the equipment described in 3.5.2 or a second identical control unit. The display shall also operate simultaneously when either one or two displays as described in 3.5.7 are added to the installation. This remote display function is not required for the C-9577A/ARC-159(V) and the C-10436A/ARC-159(V).

3.5.4.2 Controls, Radio Set C-9815/ARC-159(V), C-9816/ARC-159(V), C-9817/ARC-159(V) and C-9818/ARC-159(V) - These four control units shall provide the same functions and differ only by the panel lighting color and panel lighting voltage characteristics. The panel lighting characteristics of the control units are identified as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

Panel Lighting Voltage	Panel Lighting Color	Nomenclature
5 volts	Red	C-9815/ARC-159(V)
28 volts	Red	C-9816/ARC-159(V)
5 volts	White	C-9817/ARC-159(V)
28 volts	White	C-9818/ARC-159(V)

These control units shall meet the applicable requirements of MIL-C-6781 for Type 1 controls. The lighting of the plastic panels of the C-9815/ARC-159(V) and C-9816/ARC-159(V) shall conform to MIL-P-7788 Class 1-R (Red) Type V. The C-9817/ARC-159(V) and C-9818/ARC-159(V) shall conform to MIL-L-27160 white lighting requirements.

All other specifications are applicable to any of the four of the above listed control units.

3.5.4.2.1 Function - The requirements of 3.5.4.1.1 shall apply.

3.5.4.2.2 Form Factor - The control shall be not greater than 7.125 inches by 5.750 inches by 2.250 inches (including knobs and connectors) as shown in EI-603.

3.5.4.2.3 Weight - The weight of this control shall be not greater than 3.0 pounds.

3.5.4.2.4 Contents - The control shall be one self-contained unit suitable for cockpit mounting. It shall contain all the circuits and control facilities necessary to operate from +28 volts DC input power and to operate, from a remote location the radios and indicators of sections 3.5.2, 3.5.3, and 3.5.7.

3.5.4.2.5 Controls - Operating controls required to operate the radio set or receiver-transmitters shall be located on the front panel of the control unit. This unit shall provide the same functions as the 3.5.4.1. The control knob functions and facilities shall be as follows:

a. Function Selector Switch - This switch shall be a four position switch allowing the operator to select any of the four OFF, MAIN, BOTH, or ADF modes of operation. The respective mode characteristics shall be as defined in 3.5.1.5.1.

b. Frequency Selector Modes Switch - The frequency selector mode switch shall be a four-position switch allowing the operator to select guard, manual, preset, and preset read modes of frequency selection.

1. Guard Mode - The requirements of 3.5.1.5.2(1) shall apply except that GD shall be used to designate this position.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the current version before use.

2. Manual Mode - The requirements of 3.5.1.5.2(2) will apply except that MNL shall designate this position and the manual frequency selector controls shall be as specified in item (8) of this paragraph.

3. Preset Mode - The requirements of 3.5.1.5.2(3) shall apply except that, due to unit size constraint, an adhesive backed frequency chart separate from the unit shall be supplied.

4. Preset Read Mode - The requirements of 3.5.1.5.2(4) shall apply except that RD shall designate this position.

c. Tone Transmit Switch - The requirements of 3.5.1.5.3 shall apply.

d. Audio Volume Control - The requirements of 3.5.1.5.4 shall apply.

e. Squelch Disable Switch - The requirements of 3.5.1.5.5 shall apply.

f. Light Dimmer Control - The requirements of 3.5.1.5.6 shall apply.

g. Lamp Test Switch - The requirements of 3.5.1.5.7 shall apply.

h. Manual Frequency Select Control - Four separate three position toggle switches shall be provided on the front panel for manual selection of any of the 7000 available channels in the 225.000 through 399.975 MHz frequency range. These controls shall select the following respective frequency increments: 100 MHz and 10 MHz, 1 MHz, 0.1 MHz and 25 KHz. The mid-position of the toggle switches shall be the only "rest" position. Upward pressure shall cause the frequency to increase incrementally downward pressure shall cause the frequency to reduce incrementally.

i. Preset Channel Select Control - The requirements of 3.5.1.5.9 shall apply.

j. Load Button - A push button load switch shall be provided to transfer the selected manual frequency to the desired preset channel memory section.

3.5.4.2.6 Preset Channel Memory - A 20 channel preset memory function shall be provided within the control unit. It shall be possible to select any 20 of the 7000 channel frequencies by the 3.5.4.2.5(9). The preset memory shall be a non-destruct type such that power line interruptions or any combination of environmental conditions will not cause a loss of memory. Insertion of the desired frequencies into the respective channel sections of the memory shall be possible by operations of the front panel controls and momentarily depressing the LOAD

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the currentversion before use.

button. The frequency inserted and channel designation will be the manual frequency selected and the preset channel selected at the time the LOAD button is depressed. Since the same readout is used alternately for channel or frequency readout, either sequence of manual and preset positioning or vice versa may be used for presetting channels. The LOAD function shall be effective with the selector knob 3.5.4.2.5(2) in any position. The selected manual frequency shall remain in a held condition when the selector knob 3.5.4.2.5(2) is switched to other position and until the MNL position is reselected.

3.5.4.2.7 Frequency/Channel Display - The requirements of 3.5.1.8 shall apply except that GD, MNL, and PRESET shall indicate mode designations instead of GUARD, MANUAL, and PRESET.

3.5.4.2.8 Duty Cycle - The control unit shall be capable of continuous duty.

3.5.4.2.9 Emergency Power - The control unit shall operate with emergency input power supplied under the conditions of 3.5.1.10, 3.5.2.10 and 3.5.3.10. The control is not required to perform during power interruption but shall retain the preset channel memory information and shall return the performance as specified herein after return to applied power conditions.

3.5.4.2.10 Electrical Overload Protection - The requirements of 3.5.4.1.11 shall apply.

3.5.4.2.11 Electrical Connections - Connections to external circuits shall be provided on a receptacle as follows and shown in EI-603.

Reference Designation	Receptacle Type (or approved equiv)	Function
J1	MS 27508E14F35P	Main/Power

This power connector, even though different from that of the C-9577/ARC-159(V), shall have the same external wiring functions and characteristics as the C-9577/ARC-159(V) with one exception. 0.2 amperes of +28 volt DC primary power, not switched by the radio shall be available to this control unit in addition to the C-9577/ARC-159(V) wiring requirements. This power is required by the power control circuits to operate the on/off function.

3.5.4.2.12 Warm-Up Time - The requirements of 3.5.4.1.13 shall apply.

3.5.4.2.13 Control Operations - The control unit shall provide remote control of an equipment as described in 3.5.2 and 3.5.3 and permit all functions, capabilities and operations. For remote control of the AN/ARC-159(V)1, an external switch is required to select the unit which will control operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.4.2.14 Remote Frequency/Channel Display Operations - The requirements of 3.5.4.1.17 applicable for the C-9577/ARC-159(V) and C-10436/ARC-159(V), shall apply.

3.5.4.3 Controls, Radio Set C-9951/ARC-159(V) and C-9951A/ARC-159(V) - The controls shall be of the same design concepts and provide the same functions as the 3.5.4.1, except the dust cover connector and external wiring provision shall be configured for direct retrofit of the C-1703/ARC-27A without aircraft wiring changes. The height of this control unit is 4-7/8 inches and exceeds the height of the C-1703/ARC-27A control unit height by 3/4 inch. The functions of the C-9951/ARC-159(V) shall compare to those of the C-9577/ARC-159(V). The functions of the C-9951A/ARC-159(V) shall compare to the C-9577A/ARC-159(V).

3.5.4.3.1 Function - The control unit shall provide remote control of the RT-1150/ARC-159(V) when mounted on the MT-4659/ARC-159(V) and shall be compatible with the 3.5.7.

3.5.4.3.2 Form Factor - The control shall be not greater than 5.60 inches by 5.750 inches by 4.875 inches (including knobs and excluding the pendant connector) as shown in EI-603.

3.5.4.3.3 Weight - The weight of the control shall be not greater than 4.0 pounds.

3.5.4.3.4 Electrical Connectors - Connectors to external circuits shall be provided on a receptacle as follows and as shown in EI-603.

Reference Designation	Receptacle Type (or approved equivalent)	Function
J1	MS3106A-28-21P	Main/Power

3.5.4.3.5 Dual Remote Control Operation - Since the C-1703/ARC-27A has no switching provisions for dual remote control operation, the C-9951/ARC-159(V) and C-9951A/ARC-159(V) are not required to have this capability.

3.5.4.4 Controls, Radio Set C-9952/ARC-159(V) and C-9952A/ARC-159(V) - The controls shall be of the same design concepts and provide the same functions as the 3.5.4.1, except the dust cover, connector, and external wiring provisions shall be tailored for direct retrofit of the C-1607/ARC-52 without aircraft wiring changes. The functions of the C-9952/ARC-159(V) shall compare to those of the C-9577/ARC-159(V). The functions of the C-9952A/ARC-159(V) shall compare to the C-9577A/ARC-159(V).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.4.4.1 Function - This control unit shall provide remote control of the RT-1150/ARC-159(V) when mounted on the MT-4660/ARC-159(V) and shall be compatible with the 3.5.7.

3.5.4.4.2 Form Factor - The control shall be not greater than 6.40 inches by 5.750 inches by 4.875 inches (including knobs and integral connectors) as shown in EI-603.

3.5.4.4.3 Weight - The weight of the control unit shall be not greater than 3.60 pounds.

3.5.4.4.4 Electrical Connectors - Connections to external circuits shall be provided on a receptacle as follows and shown in EI-603.

Reference Designation	Receptacle Type (or approved equivalent)	Function
J1	K02-21-30PN	Main/Power

3.5.4.4.5 Dual Remote Control Operation - Since the C-1607/ARC-52 has no switching provisions for dual remote control operation, the C-9952/ARC-159(V) and C-9952A/ARC-159(V) are not required to have this capability.

3.5.4.5 Controls, Radio Set C-9946/ARC-159(V) and C-9946A/ARC-159(V) - The controls shall be of the same design concepts and provide the same functions as the 3.5.4.1, except the dust cover, connector, and external wiring provisions shall be tailored for direct retrofit of the C-6287/ARC-51BX, C-6476/ARC-51A, C-6555/ARC-51A or C-6556/ARC-51B without aircraft wiring changes. The functions of the C-9946/ARC-159(V) compare to those of the C-9577/ARC-159(V). The functions of the C-9946A/ARC-159(V) compare to the C-9577A/ARC-159(V).

3.5.4.5.1 Function - This control unit shall provide remote control of the RT-1150/ARC-159(V) when mounted to the MT-4661/ARC-159(V) or the RT-1194/ARC-159A(V) when mounted to the MT-4795/ARC-159A(V) and shall be compatible with the 3.5.7.

3.5.4.5.2 Form Factor - The control shall be not greater than 6.5 inches by 5.750 inches by 4.875 inches (including knobs and integral connectors) as shown in EI-603.

3.5.4.5.3 Weight - The weight of the control unit shall be not greater than 3.80 pounds.

3.5.4.5.4 Electrical Connectors - Connections to external circuits shall be provided on a receptacle as follows and shown in EI-603.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the currentversion before use.

Reference Designation	Receptacle Type (or approved equivalent)	Function
J1	MS3112E18-32P	Main/Power

3.5.4.5.5 Dual Remote Control Operation - For dual remote control operation of the receiver-transmitter, an external switch is required to select which unit will control operation.

3.5.4.6 Controls, Radio Set C-9954/ARC-159(V) and C-9954A/ARC-159(V) - The controls shall be of the same design concepts and provide the same functions as the 3.5.4.1, except the dust cover, connector and external wiring provisions shall be tailored for direct retrofit of the C-6365/ARC-34 without aircraft wiring changes. The function of the C-9954/ARC-159(V) shall compare to those of the C-9577/ARC-195(V). The functions of the C-9954A/ARC-159(V) shall compare to the C-9577A/ARC-159(V).

3.5.4.6.1 Function - This control shall provide remote control of the RT-1150/ARC-159(V) when mounted on the MT-4664/ARC-159(V) and shall be compatible with the 3.5.7 units.

3.5.4.6.2 Form Factor - The control shall be not greater than 6.20 inches by 5.750 inches by 4.875 inches (including knobs and integral connectors) as shown in EI-603.

3.5.4.6.3 Weight - The weight of the control unit shall be not greater than 3.75 pounds.

3.5.4.6.4 Electrical Connector - Connections to external circuits shall be provided on a receptacle as follows and as shown in EI-603.

Reference Designation	Receptacle Type (or approved equivalent)	Function
J1	MS-3102R-22-14P	Main/Power
J2	MS-3102R-20-29P	Indicator

3.5.5 Control, Secure Systems - The Control, Radio Set C-9451/ARC-159(V) and C-9960/ARC-159(V), when used with the appropriate switching units shall provide plain, cipher, and automatic relaying control of dual AN/ARC-159(V) systems. The controls shall meet the applicable requirements of MIL-C-6781 for Type I control panel. The lighting of the plastic panel shall conform to MIL-P-7788, Class 1-R (Navy Red) or 1-W (Air Force White in either 5 or 28 volt (AC or DC). Separate input power pins shall be provided for the 5 and 28 volt inputs such that application of power to the assigned pin provides proper illumination.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.5.1 Control Radio Set C-9451/ARC-159(V) - The control shall provide remote control of the TSEC/KY-28 secure speech equipment described in CSEEB-13 and KAM-213/TSEC plus the equipment described in 3.5.6 when interconnected for operations using two equipments described in 3.5.1, 3.5.2 or 3.5.3, with associated remote control unit. Refer to EI-603 for interconnections. For dual radio set or receiver-transmitter installations with a single TSEC/KY-28 unit and switching unit, the control shall have the capability to select either radio set for operation (normal or secure voice) with the TSEC/KY-28 or both for automatic relaying operations (normal or secure voice).

3.5.5.1.1 Form Factor - The control shall be not greater than 4.25 inches by 5.75 inches by 1.50 inches (including knobs and integral connector) as shown in EI-603.

3.5.5.1.2 Weight - The weight of the control shall be not greater than 1.0 pounds.

3.5.5.1.3 Contents - The control shall be one self-contained unit suitable for cockpit mounting and containing switches to control operation of the TSEC/KY-28 plus equipment described in 3.5.6.1.

3.5.5.1.4 Controls - Controls required for operation of the TSEC/KY-28 and switching unit shall be located on the front panel. All control positions and resulting operations are described for interconnection with TSEC/KY-28, the switching unit, and two radio sets.

3.5.5.1.4.1 TSEC/KY-28 Mode Selector Switch - This switch shall control the mode of TSEC/KY-28 operation as follows:

a. Plain Mode of OFF Mode - In this position, labeled P/OFF, the 28 volts DC input power shall be switched to the TSEC/KY-28 receive plain/cipher control input. While in this position, 28 volts DC input power shall also be switched to the switching unit if the radio mode selector switch is in the RELAY position.

b. Cipher Mode - In this position, labeled C, 28 volts DC input power shall be applied to the TSEC/KY-28 main power input and an electrical ground applied to its receive plain/cipher control input. While in this position, the TSEC/KY-28 shall be switched to the radio or for relay operations as determined by the position of the radio mode selector switch. Input power shall be maintained to the switching unit.

c. Retransmit Delay Mode - In this position, labeled DELAY, an electrical ground shall be applied to the TSEC/KY-28 external time delay control input circuit while maintaining 28 volt DC input power to the TSEC/KY-28 and the switching unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.5.1.4.2 Radio Mode Selector Switch - This switch shall select automatic relaying mode or which of two radio sets will be connected to a single TSEC/KY-28 for secure voice operation.

a. Radio Number One (1) Secure Mode - In this position labeled RAD-1, the single TSEC/KY-28 shall be connected to radio number one (1) if the TSEC/KY-28 Mode Selector Switch is in any position except P/OFF. In this position, radio number one (1) shall be connected to the TSEC/KY-28 if the TSEC/KY-28 Mode Selector Switch is in the P/OFF position. For either case, radio number two (2) shall not be connected to the TSEC/KY-28.

b. Radio Number Two (2) Secure Mode - In this position, labeled RAD-2, the single TSEC/KY-28 shall be connected to radio number two (2) if the TSEC/KY-28 Mode Selector Switch is in any position except P/OFF. In this position radio number two (2) shall not be connected to the TSEC/KY-28 if the TSEC/KY-28 Mode Selector Switch is in the P/OFF position. For either case, radio number one (1) shall not be connected to the TSEC/KY-28.

c. Automatic Relay Mode - In this position, labeled RELAY, the two radio sets or receiver-transmitters shall be interconnected by the switching unit for automatic relaying operations described in 3.5.1.19, 3.5.2.19, and 3.5.3.19. In this position, normal voice signals shall be relayed if the TSEC/KY-28 Mode Selector Switch is in the P/OFF position and secure voice signals shall be relayed if it is in any other position. In this position and when relaying secure voice signals, the TSEC/KY-28 shall be connected through the switching unit to decipher (monitor) signals received on either radio set or receiver-transmitter. In this position, actuation of either radio set or receiver-transmitter external transmit (key) switch shall cause both to transmit the same signals simultaneously. The signals to be transmitted shall pass through the TSEC/KY-28 prior to transmission for all positions of the TSEC/KY-28 Mode Selector Switch except P/OFF.

3.5.5.1.4.3 TSEC/KY-28 Zeroize Switch - This switch, labeled ZEROIZE, shall connect +28 volts DC to the TSEC/KY-28 zeroize control input when actuated. The switch shall have a protective feature to prohibit accidental actuation while not requiring extraordinary efforts to actuate whn desired. The switch shall be capable of conducting 7 amperes surge current for 15 milliseconds.

3.5.5.1.5 Duty Cycle - The control shall be capable of continuous operation under standard and service conditions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.5.1.6 Emergency Power - The control unit shall operate from +28 volts DC primary power inputs. Under emergency power conditions, the switches contained in the control shall continue to provide all required power switching capabilities.

3.5.5.1.7 Electrical Overload Protection - Electrical overload protection and fuses are not required in the control unit.

3.5.5.1.8 Electrical Connections - Connections to external circuits shall be provided on a receptacle as follows and shown in EI-603.

Reference Designation	Receptacle Type (or approved equivalent)	Function
J1	MS27508E-12F-35P	Main/Power

3.5.5.1.9 Warm-Up Time - The time required for the control unit to warm-up prior to operation shall be not greater than 1.0 second and shall result in switching unit operation as specified in 3.5.6.1.10.

3.5.5.2 Control, Radio Set C-9960/ARC-159(V) - The control shall provide remote control of the TSEC/KY-28 secure voice equipments described in CSEEB-13 and KAM-213/TSEC plus the equipment described in 3.5.6.2 when interconnected for operations using two equipments described in 3.5.1, 3.5.2 and 3.5.3, with associated control units. Interconnections are identified in EI-603. With the system of two radios, two TSEC/KY-28 units and the SA-2078/ARC-159(V), the control shall have capability to designate normal or secure voice operation of the radios and automatic relaying in either plain or secure mode.

3.5.5.2.1 Form Factor - The control shall be not greater than 4.25 inches by 5.750 inches by 1.50 inches (including knobs and connector) as shown in EI-603.

3.5.5.2.2 Weight - The weight of the control shall be not greater than 1.50 pounds.

3.5.5.2.3 Contents - The control shall be one self contained unit suitable for cockpit mounting and containing switches to control operation of the TSEC/KY-28 units plus equipment described in 3.5.8.2.

3.5.5.2.4 Controls - Controls required for operation of the TSEC/KY-28 units and switching unit shall be located on the front panel. All control positions and resulting operations are described with the control interconnected in a system of the above units and two radio sets.

3.5.5.2.4.1 TSEC/KY-28 Mode Selector Switch - This six position switch shall select the mode of operation as follows:

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the currentversion before use.

a. Plain Mode or OFF Mode - In this position, labeled OFF, the two radio sets shall operate as independent radios and shall both be in plain voice mode.

b. CRYPTO 1 - In this position, the two radio sets shall operate as independent radios. The number one radio, coupled to the number one TSEC/KY-28, shall be in secure voice mode and the number two radio shall be in plain voice mode.

c. CRYPTO 2 - In this position, the two radio sets shall operate as independent radios. The number one radio shall be in plain voice mode and the number two radio, coupled to the number two TSEC/KY-28 shall be in secure voice mode.

d. CRYPTO 1 & 2 - In this position, the two radio sets shall operate as independent radios. Each radio coupled to the respective TSEC/KY-28 shall be in secure voice mode.

e. CRYPTO RELAY - In this position, the two radios shall be coupled through the SA-2078/ARC-159(V) for automatic relaying operation in secure voice mode as described in 3.5.1.19, 3.5.2.19, 3.5.3.19 and 3.5.3.2.14. The incoming demodulated signal shall be coupled through the switching unit to the TSEC/KY-28 to decipher (monitor) signals received by either radio set or receiver-transmitter. In this position, actuation of either radio set or receiver-transmitter external transmit (key) switch shall cause both transmitters to transmit simultaneously. The modulation signals to be transmitted shall be connected through the TSEC/KY-28 equipments for processing prior to transmission.

f. Plain Relay - In this position, the two radios shall be coupled through the SA-2078/ARC-159(V) for automatic relaying of plain voice modulation as described in 3.5.1.19, 3.5.2.19, and 3.5.3.1.19. The incoming demodulated signals shall be available to the radio set operators for monitoring. In this position, actuation of either radio set or receiver-transmitter transmit (key) switch shall cause both radios to transmit simultaneously. The keying operators microphone output shall be connected to the modulator input of both radios for simultaneous modulation of the transmitters.

3.5.5.2.4.2 Delay Switch - The C-9960/ARC-159(V) shall have a two position switch for control of the TSEC/KY-28 delay function. When in DELAY position an electrical ground shall be applied to the two TSEC/KY-28 equipments to activate the delay circuits.

3.5.5.2.4.3 Zeroize Switch - The C-9960/ARC-159(V) shall have a switch labeled ZEROIZE that when actuated shall connect +28 volts DC to the TSEC/KY-28 zeroize control circuits. This switch shall have a protective feature to prohibit accidental actuation while not requiring extraordinary efforts to actuate when desired. The switch shall be capable of conducting 14 ampere surge current for 15 milliseconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.5.2.5 Duty Cycle - The control shall be capable of continuous operation under standard or service conditions.

3.5.5.2.6 Emergency Power - The requirements of 3.5.5.1.7 shall apply.

3.5.5.2.7 Electrical Overload Protection - Electrical overload protection and fuses are not required in the control unit.

3.5.5.2.8 Electrical Connections - Connections to external circuits shall be provided on a receptacle as follows and shown in EI-603.

Reference Designation	Receptacle Type (or approved equivalent)	Function
J1	MS27508E-12F-35P	Main/Power

3.5.5.2.9 Warm-Up Time - The time required for the control unit to warm-up prior to operation shall be not greater than 1.0 second and shall result in switching unit operations as specified in 3.5.8.2.10.

3.5.6 Switching Units - The switching units SA-1964/ARC-159(V) and SA-2078/ARC-159(V) when operated in conjunction with their respective companion control units shall provide the system interface switching functions for the designated operational modes.

3.5.6.1 Switching Unit SA-1964/ARC-159(V) - The switching unit shall operate in conjunction with the equipment described in 3.5.5.

3.5.6.1.1 Function - The switching unit shall operate with its associated control unit to provide switching and interface of signals when interconnected for operations using a TSEC/KY-28 plus two equipments described in 3.5.1, 3.5.2, or 3.5.3, with associated control unit. Refer to EI-603 for interconnections. Using these connections, the switching units shall transfer signals between either radio set for operation (normal or secure voice) with the TSEC/KY-28 or between both for automatic relaying operations (normal or secure voice).

3.5.6.1.2 Form Factor - The switching unit shall be not greater than 7.75 inches by 3.10 inches by 3.10 inches (including mounting flanges and connector) as shown in EI-603.

3.5.6.1.3 Weight - The weight of the switching unit shall be not greater than 2.1 pounds.

3.5.6.1.4 Contents - The switching unit shall be one self-contained unit suitable for mounting outside the cockpit area. It shall contain a power supply and relay circuits for switching incoming audio and control signals to provide the operational modes selected by the C-9451/ARC-159(V) control as described in 3.5.5.1.4, except for the ZEROIZE

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

function. Additionally, the switching unit shall contain two sensing and two amplifier circuits for use in the automatic secure voice relay operations. The sensing circuits shall automatically determine when an incoming secure voice signal is present at either of the two associated radio sets and cause the other to transmit the secure voice signal. The amplifier circuits shall separately amplify signals from their respective associated radio TSEC/KY-28 wideband audio output to the level required to modulate the other radio set to between 90 and 100 percent under standard conditions.

3.5.6.1.5 Controls - No controls are required on the switching unit. All control functions are provided by connection to the equipment described in 3.5.5.

3.5.6.1.6 Duty Cyle - The switching unit shall be capable of continuous operation under standard and environmental conditions as specified in MIL-E-5400 for Class II equipment.

3.5.6.1.7 Emergency Power - The switching unit shall operate from 28 volts DC primary power inputs. Under emergency power conditions, the switching unit shall not be required to function in the automatic relay (normal or secure voice) modes. However, the transmit precedence feature of the automatic relay (normal and secure voice) mode shall be operative, whereby actuation of either radio set or receiver-transmitter external transmit (key) switch shall cause both to transmit the same signals simultaneously.

3.5.6.1.8 Electrical Overload Protection - The switching unit shall be self protecting against overload (above reserve capability) due to electrical faults and overloads within the equipment. The switing unit shall be provided with overload protection by means of a 1/2 amp. fuse conforming to the requirements of MIL-F-15160. The fuse shall be readily accessible with a spare fuse provided in the equipment. Only one fuse shall be used in the plus (+) DC input power line as close as possible to the input power connector.

3.5.6.1.9 Electrical Connections - Connections to external circuits shall be provided on a receptacle as follows and shown in EI-603.

Reference Designation	Receptacle Type (or approved equivalent)	Function
J1	MS3112E-22-55P	Main/Power

3.5.6.1.10 Warm-Up Time - The time required for the switching unit to warm-up to operation shall be not greater than 1.0 second.

3.5.6.2 Switching Unit SA-2078/ARC-159(V) - The switching unit shall operate in conjunction with the equipment described in 3.5.5.2.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.6.2.1 Function - The requirements of 3.5.8.1.1 shall apply except that the system shall include two TSEC/KY-28 equipments instead of one.

3.5.6.2.2 Form Factor - The form factor shall be the same as that of the SA-1964/ARC-159(V) 3.5.8.1.2.

3.5.6.2.3 Weight - The weight of the switching unit shall be not greater than 2.50 pounds.

3.5.6.2.4 Contents - The switching unit shall be one self-contained unit suitable for mounting outside the cockpit area. It shall contain a power supply and relay circuits for switcing incoming signals to provide the operational modes selected by the 3.5.5.2 as described in 3.5.5.2.5. Additionally, the switching unit shall contain two sensing and two amplifier circuits for use in the automatic secure voice relay operations. The sensing circuits shall automatically determine when an incoming secure voice signal is present at either of the two associated radio sets and cause the other to transmit the secure voice signal. The amplifier circuits shall separately amplify signals from their associated radio TSEC/KY-28 wideband audio output to the level required to modulate the other radio set to between 90 and 100 percent under standard conditions.

3.5.6.2.5 Controls - No controls are required on the switching unit. All control functions are provided by connection to the equipment described in 3.5.5.2.

3.5.6.2.6 Duty Cycle - The requirements of 3.5.6.1.6 shall apply.

3.5.6.2.7 Emergency Power - The requirements of 3.5.6.1.7 shall apply.

3.5.6.2.8 Electrical Overload Protection - The requirements of 3.5.6.1.8 shall apply.

3.5.6.2.9 Electrical Connections - The requirements of 3.5.6.1.9 shall apply.

3.5.6.2.10 Warm-Up Time - The requirements of 3.5.6.1.10 shall apply.

3.5.7 Indicators, Frequency-Channel - The ID-1972/ARC-159(V) and ID-1984/ARC-159(V) shall be one package units suitable for cockpit instrument panel installation. Either of these indicator units shall present duplicate Frequency-Channel readout information of that shown by the active system control unit 3.5.4.1 or equivalent. The Indicator Group OD-122/ARC-159(V) shall provide the same functions and displays as the one package indicators except it shall consist of two units, the ID-2053/ARC-159(V) and the CV-3328/ARC-159(V).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.7.1 Indicator, Frequency-Channel ID-1972/ARC-159(V) - The indicator shall meet the applicable requirements of MIL-E-5400 for cockpit mounted equipment. The panel lighting color shall conform to MIL-P-7788 Class 1-R (Navy Red) or 1-W (Air Force White in accordance with MIL-L-27160) with a lamp voltage input of either 5 or 28 volts (AC or DC). Separate input power pins shall be provided for the 5 and 28 volt inputs such that application of power to the assigned pin provides proper illumination.

3.5.7.1.1 Function - The indicator shall provide remote display of the operating radio frequency or preset channel number selected by the equipment described in 3.5.2 and 3.5.4.

3.5.7.1.2 Form Factor - The indicator shall be of a rectangular design with dimensions shall be not greater than 7.35 inches by 3.30 inches by 1.325 inches (including knobs and connector) as shown in EI-603.

3.5.7.1.3 Weight - The weight of the indicator shall be not greater than 1.1 pounds.

3.5.7.1.4 Contents - The indicator shall be one self-contained unit suitable for cockpit mounting, and contain the following assemblies:

a. Frequency/Channel Display Assembly -

b. Power Supply/Chassis Assembly -

c. Converter Board Assembly -

The frequency/channel display shall be six digits visible through a red filter as described in 3.5.1.8. The power supply/chassis assembly shall include the main chassis structure plus a power supply for converting the 28 volt DC input power to that required by the display circuitry. The converter board shall contain circuitry for conversion of the serial digit input to parallel coded output for driving the frequency/channel display digits. The converter shall operate from the digital serial information supplied through two pairs of wires from the equipment described in 3.5.2 or 3.5.4. All other indicator functions shall be accommodated by conventional wiring for the respective functions.

3.5.6.1.5 Controls - Front panel controls shall consist of a dimmer control and lamp test switch. The dimmer control shall vary the illumination intensity of the incandescent lamps of the frequency/channel display by varying their input voltage downward from the maximum rated voltage. Activation of the lamp test switch shall illuminate all the elements of seven segment units in all six digits. A power on/off switch is not required since input power is supplied and controlled by the associated radio system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.7.1.6 Duty Cycle - The indicator shall be capable of continuous operation under either standard or service conditions.

3.5.7.1.7 Emergency Power - The indicator shall operate from +28 volts DC primary power inputs from the associated radio system. Under emergency power conditions, the indicator shall display the proper frequency or channel numbers.

3.5.7.1.8 Electrical Overload Protection - Electrical overload protection and fuses are not required in the indicator.

3.5.7.1.9 Electrical Connections - Connections to external circuits shall be provided on a receptacle as follows and shown in EI-603.

Reference Designation	Receptacle Type (or approved equivalent)	Function
J1	MS27508E-10F-35P	Main/Power

3.5.7.1.10 Warm-Up Time - The time required for the indicator to warm-up prior to operation shall be not greater than 1.0 second.

3.5.7.1.11 Accuracy - The indicator shall display the same operating frequency or preset channel as displayed on the associated radio set or remote control.

3.5.7.2 Indicator, Frequency-Channel ID-1984/ARC-159(V) - The requirements of 3.5.7.1 shall apply unless otherwise specified herein.

3.5.7.2.1 Function - The requirements of 3.5.7.1.1 shall apply.

3.5.7.2.2 Form Factor - The indicator shall be of a cylindrical configuration as shown in EI-603. Maximum dimensions shall be 7.3 inches length by 1.94 inches body and 2.005 inches panel diameters. The front section configuration shall permit attachment of a mounting flange having outline configuration and mounting holes defined by MS-28055-6.

3.5.7.2.3 Weight - The weight of the indicator shall be not greater than 1.1 pounds.

3.5.7.2.4 Contents - The indicator shall be one self-contained unit suitable for cockpit mounting, and contain the following assemblies:

a. Frequency/Channel Display Assembly
b. Power Supply Board
c. Data Converter Board
d. Decoder/Driver Board

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

The frequency/channel display shall be six digits visible through a red filter as described in 3.5.1.8 except that the digits shall be arranged in two horizontal rows with four most significant digits in the upper row and the two least significant digits in the lower row. The power supply board shall contain circuitry for converting the 28 volt DC input power to that required by the display and logic circuitry. The data converter and decoder/driver boards shall contain circuitry for conversion of the serial digital input to a parallel format for driving the display and circuitry for varying the intensity of the incandescent display. The converter shall operate from the digital serial information supplied through two pairs of wires from the equipment described in 3.5.2 or 3.5.4. All other indicator functions shall be accommodated by conventional wiring for the respective functions.

3.5.7.2.5 Controls - Front panel controls shall consist of one control which shall vary display intensity and control the lamp test function. Turning the control clockwise shall increase the illumination intensity of the incandescent frequency channel display. Turning the control fully clockwise shall illuminate all seven segment units in all six digits at maximum intensity. A power on/off switch is not required since input power is supplied and controlled by the associated radio system.

3.5.6.2.6 Duty Cycle - The requirements of 3.5.7.1.6 shall apply.

3.5.7.2.7 Emergency Power - The requirements of 3.5.7.1.7 shall apply.

3.5.7.2.8 Electrical Overload Protection - The requirements of 3.5.7.1.8 shall apply.

3.5.7.2.9 Electrical Connections - The requirements of 3.5.7.1.9 shall apply.

3.5.7.2.10 Warm-Up Time - The requirements of 3.5.7.1.10 shall apply.

3.5.7.2.11 Accuracy - The requirements of 3.5.7.1.11 shall apply.

3.5.7.3 Indicator Group OD-122/ARC-159(V) - The indicator group, to minimize instrument panel space required, shall consist of two units, the Indicator Frequency/Channel ID-2053/ARC-159(V) and the converter, digital CV-3328/ARC-159(V). The system shall function normally with the readout unit ID-2053/ARC-159(V) located in the instrument panel area and the converter CV-3328/ARC-159(V) located elsewhere in the aircraft.

3.5.7.3.1 Indicator Unit ID-2053/ARC-159(V) - The requirements of 3.5.7.1 shall apply unless otherwise specified herein.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.7.3.1.1 Function - The requirements of 3.5.7.1.1 shall apply.

3.5.7.3.1.2 Form Factor - The indicator shall be of a rectangular design except the rear portion of the top surface shall be sloping. The dimensions shall be not greater than 3.23 inches by 1.88 inches by 1.59 inches (not including knob) as shown in EI-603.

3.5.7.3.1.3 Weight - The weight of the indicator shall be not greater than 0.4 pounds.

3.5.7.3.1.4 Contents - The indicator unit shall be suitable for cockpit mounting in special applications and shall contain the following assemblies:

a. Frequency/Channel Display Assembly
b. Upper Logic Board
c. Lower Logic Board

The frequency/channel display shall be six digits visible through a red filter as described in 3.5.1.8 except that the digits shall be arranged in two horizontal rows of three digits each. The two logic boards shall contain circuitry for converting the serial digital data from the CV-3328/ARC-159(V) to a parallel format and driver circuitry for the incandescent display.

3.5.7.3.1.5 Controls - Front panel controls shall consist of one control which shall vary display intensity and control the lamp test function. Turning the control clockwise shall increase the illumination intensity of the incandescent frequency/channel display. Turning the control fully clockwise shall illuminate all seven segment units in all six digits at maximum intensity. A power on/off switch is not required since input power is supplied and controlled by the associated radio system.

3.5.7.3.1.6 Duty Cycle - The requirements of 3.5.7.1.6 shall apply.

3.5.7.3.1.7 Emergency Power - The requirements of 3.5.7.1.7 shall apply.

3.5.7.3.1.8 Electrical Overload Protection - The requirements of 3.5.7.1.8 shall apply.

3.5.7.3.1.9 Electrical Connections - Connections to external circuits shall be provided on a receptacle as follows and shown in EI-603.

Reference Designation	Receptacle Type (or approved equivalent)	Function
J1	Continental MM14-22PGDFS	Main/Power

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.7.3.1.10 Warm-Up Time - The requirements of 3.5.7.1.10 shall apply.

3.5.7.3.1.11 Accuracy - The requirements of 3.5.7.1.11 shall apply.

3.5.7.3.2 Power-Converter Unit CV-3328/ARC-159(V) - The power converter unit, when used in conjunction with 3.5.7.1 shall meet the requirements of 3.5.6.1 unless otherwise specified herein.

3.5.7.3.2.1 Function - The power-converter unit shall provide DC voltage and conditioned digital signals as required by 3.5.7.3.1.

3.5.7.3.2.2 Form Factor - The power-converter unit shall be not greater than 5.36 inches by 3.56 inches by 1.65 inches (including mounting flanges and connectors) as shown in EI-603.

3.5.7.3.2.3 Weight - The weight of the power-converter unit shall be not greater than 0.9 pounds.

3.5.7.3.2.4 Contents - The power-converter unit shall be one self-contained unit suitable for mounting outside the cockpit area. It shall contain circuitry for providing necessary digital signals +5 and -12 volt DC outputs and a variable DC output in response to a control voltage from 3.5.7.3.1.

3.5.7.3.2.5 Controls - No controls are required on the power-converter unit.

3.5.7.3.2.6 Duty Cycle - The requirements of 3.5.7.1.6 shall apply.

3.5.7.3.2.7 Emergency Power - The requirements of 3.5.7.1.7 shall apply.

3.5.7.3.2.8 Electrical Overload Protection - The requirements of 3.5.7.1.8 shall apply.

3.5.7.3.2.9 Electrical Connections - Connections to external circuits shall be provided on receptacles as follows and shown in EI-603.

Reference Designation	Receptacle Type (or approved equivalent)	Function
J1	MS27508E-10F-35P	Main/Power Interface with 3.5.7.1 Indicator Unit, ID-2053/ARC-159(V)
J2	MS3112E14-19P	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.7.3.2.10 Warm-Up Time - The requirements of 3.5.7.1.10 shall apply.

3.5.8 Filter Converter F-1420/ARC-159A(V) - This filter converter shall operate with the AN/ARC-159A(V) system, normally mounted to the MT-4795/ARC-159A(V) and shall provide immunity to the RT-1194/ARC-159A(V) from off-channel frequency signals in both receiver and transmit operation.

3.5.8.1 Contents - The filter-converter shall be one self-contained unit and contains the following functional circuits and assemblies:

a. UHF Provisions

The filter-converter shall provide frequency selectivity and shall have coaxial connections for and operate in the normal 50 ohm impedance line between the RT-1194/ARC-159A(V) receiver-transmitter and the antenna.

b. Tuning Provisions

The filter-converter shall have connector provisions, circuits, and assemblies to operate from the RT-1194/ARC-159A(V) system frequency selection information and automatically tune the filter to the selected channel frequency.

c. Power Supply

The filter-converter shall have power supply accommodations to operate from 27.5 volt DC power.

d. Guard Channel Frequency Acceptance

The filter-converter shall have bypass circuitry to accept the guard channel frequency, separate from the main channel frequency selected. A separate guard channel coaxial connector shall not be used.

e. Guard Disconnect Operation

The guard amplifier shall be disabled by the guard disconnect control bit of the serial bit stream. During this mode of operation, the transmit loss specification shall be applicable on the main channel and the guard isolation specification applicable on the guard output.

3.5.8.2 Operational Characteristics - The F-1420/ARC-159A(V) shall operate simultaneously with the RT-1194/ARC-159A(V) and shall automatically tune to the frequency selected by the system control unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.8.2.1 UHF Characteristics - The filter shall provide selectivity at the main receiver-transmitter frequency. The filter shall include bypassing provisions for the guard channel frequency and then circuits to recombine the main and guard frequencies in the receive mode. The main channel bandpass filter section shall tune over the 225 MHz through 399.9 MHz range and the guard bypass section shall be at a fixed frequency, normally 243 MHz, but adjustable over the range of 238 MHz to 248 MHz.

3.5.8.2.1.1 Selectivity - The selectivity characteristics for the transmitter, main receiver and guard frequencies shall be as follows:

3.5.8.2.1.1.1 Selectivity-Transmit - The main channel attenuation of frequencies 10 MHz or more from the selected frequency shall be at least 45 dB under standard conditions and 40 dB under service conditions.

3.5.8.2.1.1.2 Selectivity-Receiver - The selectivity will be affected by the recombining of the main and guard frequencies. Above 310 MHz the frequencies shall be independent such that attenuation 10 MHz or more from the selected frequency is at least 45 dB under standard conditions and 40 dB under service conditions.

3.5.8.2.1.1.3 Selectivity-Guard - The guard channel attenuation of frequencies 15 MHz or more from the guard frequency shall be at least 30 dB when the main receiver is at least 25 MHz away from the guard frequency.

3.5.8.2.1.2 Insertion Loss - The UHF insertion loss shall be not greater than the following:

Transmit: Standard conditions, 1.3 dB
Service conditions, 1.5 dB

Receive: (Main Channel)
Standard conditions, 1.8 dB
Service conditions, 2.0 dB

NOTE: Frequencies within ±10 MHz of the guard frequency are excluded from these requirements.

Standard and service conditions, 6 dB maximum, when the main receiver is tuned at least 10 MHz away from the guard frequency.

3.5.8.2.1.3 Voltage Standing Wave Ratio (VSWR) - The VSWR looking into the UHF port with the antenna port terminated into a 50 ohm resistive load shall be:

1.6:1 maximum under standard conditions
2.0:1 maximum under service conditions

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.8.2.1.4 RF Power Capability - The main channel shall handle a power level of 30 watts amplitude modulated carrier at 95 percent modulation.

3.5.8.2.2 Tuning - The F-1420/ARC-159A(V) shall operate from the tuning information supplied to the RT-1194/ARC-159A(V) and by servo tuning or equivalent shall tune to the system frequency selected.

3.5.8.2.2.1 Time - The nominal tuning time (225 to 399.9 MHz) shall be not greater than 3 seconds. Under service conditions, the tuning time shall be not greater than 5 seconds.

3.5.8.2.2.2 Transmit Inhibit - While the filter tuning is in progress, transmission shall be automatically inhibited by actuation of the transceiver key inhibit line.

3.5.8.2.2.3 Serial Control - The filter-converter shall operate from the serial frequency control information generated by the AN/ARC-159A(V) control system. The input impedance of the control circuits shall be 40,000 ohms or greater and shall be balanced. The converter tuning system shall recognize and tune in accordance with the incremental 100 MHz, 10 MHz, 1 MHz and 100 KHz serial digital information supplied to it. The system 50 KHz and 25 KHz information need not be recognized by the filter-converter. The bandpass frequency shall be sufficient to cover the 50 KHz and 25 KHz increments without retuning. In addition, the filter-converter shall recognize the guard disconnect bit (Control Bit B) and control the guard preamplifier in the filter in accordance with 3.5.8.1(5).

3.5.8.3 Environmental Capability - The filter shall be capable of continuous rf operation over all environmental conditions, specified for the AN/ARC-159A(V) system. These include temperature altitude per MIL-E-5400 Class 2, Curve A, and vibration per MIL-E-5400 Curve III solid mounted and Curve IV mounted on the MT-4795/ARC-159A(V) or MT-4840/ARC-159A(V).

3.5.8.4 Construction - The filter unit shall be a sealed, pressurized assembly to allow full power operation at 70 K ft. altitude. The chassis shall consist of a lower case which contains four foreshortened quarter-wave resonators and a cover containing a sealed gasket. Both the case and cover shall be fabricated from aluminum to the maximum extent possible to optimize strength, weight, and frequency stability. Mechanical drive to the resonator tuning capacitors shall be obtained by a drive assembly located within the sealed enclosure. Associated control and power supply circuits are constructed under removable covers and shall be outside of the pressurized area.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.8.5 Power-Requirements - The filter-converter shall operate satisfactorily from MIL-STD-704 category B DC power under the various operating conditions including emergency conditions. During standby, (tuning quiescent) the line current shall not exceed 1.1A at 27.5 VDC. With tuning in progress, the line current shall be not greater than 1.6A at 27.5 VDC.

3.5.8.6 Form Factor - The antenna filter shall be not greater than 14.32 inches depth by 2.67 inches width by 6.50 inches height (including connector and mounting projections as shown in EI-603).

3.5.8.7 Weight - The weight of the filter-converter shall be not greater than 8.0 pounds.

3.5.8.8 Electrical Connections - Connections to external circuits shall be provided as follows and shown in EI-603.

Reference	Connector Type (or approved equivalent)	Function
J1	Coaxial Type N	Antenna
J2	Bendix 21-349214-18P	Power/Control
J3	Coaxial Type N	R/T RF

3.5.9 Power Supplies - Ancilliary power supply units for AN/ARC-159 and AN/ARC-159(V) systems installations shall provide the power conversions as defined herein.

3.5.9.1 Power Supply PP-7095/ARC-159(V) - The power supply shall work in conjunction with equipment described in 3.5.1 and 3.5.3, with associated control to provide +225 volt DC power required for operation AN/ARA-25 Direction Finder Group.

3.5.9.1.1 Function - The power supply shall operate to provide a +225 volt DC ±10 percent output with a load requiring 35 milliamperes when interconnected for operations using the AN/ARA-25 and associated radio sets as shown in EI-603.

3.5.9.1.2 Form Factor - The power supply shall be not greater than 4.50 inches by 2.5 inches by 2.0 inches (including mounting flanges and connector) as shown in EI-603.

3.5.9.1.3 Weight - The weight of the power supply shall be not greater than 1.0 pound.

3.5.9.1.4 Contents - The power supply shall be one self-contained unit suitable for mounting outside the cockpit area. It shall contain conventional solid state power supply circuitry and components.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.9.1.5 Controls - No controls are required on the power supply unit. The output voltage shall be supplied automatically upon application of +28 volt DC input power from the associated radio set.

3.5.9.1.6 Duty Cycle - The power supply shall be capable of continuous operation under standard and service conditions.

3.5.9.1.7 Emergency Power - Operation under emergency power conditions is not required. However, operation is required for input power down to +22 volts DC.

3.5.9.1.8 Electrical Overload Protection - Electrical overload shall be provided on a receptacle as follows and shown in EI-603.

Reference Designation	Receptacle Type (or approved equivalent)	Function
J1	MS3110E-8-4	Main/Power

3.5.9.1.10 Warm-Up Time - The time required for the power supply to warm-up prior to operation shall be not greater than 1.0 second under standard and service conditions.

3.5.9.2 Converter, Power Supply, PP-7323/ARC-159A(V)-

3.5.9.2.1 Function - This converter power supply shall convert 115 volt 400 Hz AC primary power to 27.5 volt DC power for a portion of the primary power to the RT-1194/ARC-159A(V). The converter shall operate satisfactorily within the environmental conditions of MIL-E-5400 Class I conditions.

3.5.9.2.2 Form Factor and Mounting - The maximum overall dimensions of the PP-7323/ARC-159A(V) shall be 6.75 inches by 4.75 inches by 1.5 inches with configuration and mounting as defined in EI-603. The converter shall be compatible with the mounting provisions in the MT-4795/ARC-159A(V) and shall have a pendant cable for connection to the connector also provided in the mounting.

3.5.9.2.3 Primary Power - The converter shall operate from 115 volt 400 Hz, 3 phase AC primary power per MIL-STD-704 Category B and shall provide 27.5 volt DC power out within the limits of MIL-STD-704 Category B. A minimum of 3.2 amperes power at 27.5 volts DC shall be provided with 115 volt AC input. Input power required under 115 volt AC input conditions and 3.2 amperes output shall not exceed 140 watts.

3.5.9.2.4 Weight - The weight of this unit shall be not greater than 3.5 pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.9.2.5 Contents - The converter shall contain solid state circuitry and components as appropriate for converting three phase 115 volt 400 Hz power to 27.5 volt DC power. It shall also contain filtering as necessary for electromagnetic interference immunity. Power on/off switching is not required within this unit.

3.5.9.2.6 Vibration - This converter shall be subjected to vibration only when attached to the MT-4795/ARC-159A(V).

3.5.9.2.7 Electrical Connections - The converter shall include a pendant cable and connector as follows and wired as shown in EI-603.

Reference Designation	Receptacle Type (or approved equivalent)	Function
J1	MS-3116E-14-19P	Main/Power

3.5.10 HD-1001/ARC-159A(V) Cooler, Air, Electronic Equipment -

3.5.10.1 Function - The cooler shall provide cooling air to the RT-1194/ARC-159A(V) when the radio set is operating in a self-cooling mode. The cooler shall have capability for providing the thermal needs of the receiver-transmitter under any temperature-altitude combination in the range of MIL-E-5400 Class 1 limits.

3.5.10.2 Form Factor and Mounting - The cooler shall be a single subassembly capable of being attached to the rear of the MT-4795/ARC-159A(V) by means of appropriate screws. It shall be possible to remove the cooler from the mounting by use of ordinary hand tools. The cooler shall have a pendant cable for connecting to the power connector provided on the MT-4795/ARC-159A(V). Maximum overall dimensions of the cooler shall be 5.5 inches by 4.5 inches by 3.0 inches. Configuration of the HD-1001/ARC-159A(V) shall be as defined in EI-603.

3.5.10.3 Primary Power - The cooler shall operate from 115 volt 400 Hz single phase power in accordance with MIL-STD-704 Category B power under all conditions including emergency levels. Primary power input shall be 25 watts maximum.

3.5.10.4 Weight - The weight of this unit shall be not greater than 1.1 pounds.

3.5.10.5 Contents - This cooler shall contain a blower and air transition means for circulating ambient air through the receiver-transmitter. The air flow and pressure rates shall be compatible with the characteristics defined in 3.5.3.2.5. Air flow of the cooler unit not attached to a receiver/transmitter or other unit creating back pressure shall be:

a. Sea level pressure, 55 pounds per hour minimum
b. 50,000 foot altitude, 44 pounds per hour minimum

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.10.6 Vibration - The cooler shall be subjected to vibration only when attached to the MT-4795/ARC-159A(V).

3.5.10.7 Electrical Connections - The cooler shall include a pendant cable and connector as follows and wired as shown in EI-603.

Reference Designation	Receptacle Type (or approved equivalent)	Function
J1	MS-3116F-10-6P	Main/Power

3.5.11 Mounting Base, Electrical Equipment MT-4609/ARC-159(V) - The mounting base shall be designed to secure the SA-1964/ARC-159(V) or SA-2078/ARC-159(V) switching unit to an airframe structure outside the cockpit area. The mount shall have provisions for solidly securing it to the airframe structure. It shall provide a grooved section at the rear for engaging the switching unit mounting flange and positive self-locking thumbscrew fasteners at the front for securing the unit to the mount. The mount shall meet the crash safety requirements and vibration per Curv IV of MIL-E-5400, and shall not require vibration isolators.

3.5.11.1 Form Factor - The mount shall be not greater than 10.5 inches by 4.5 inches by 2.5 inches (including flanges and thumbscrew fasteners) as shown in EI-603.

3.5.11.2 Weight - The weight of the mount shall be not greater than 0.5 pound.

3.5.12 Mounting Base-AN/ARC-159(V)2 Receiver-Transmitter - Each of the following mounting bases shall secure the RT-1150/ARC-159(V) of an AN/ARC-159(V)2 system to the aircraft.

3.5.12.1 Mounting Base, Electrical Equipment MT-4658/ARC-159(V) - The mounting base shall be designed to secure the RT-1150/ARC-159(V) receiver-transmitter to an airframe structure outside the cockpit area. The mount shall have provisions for solidly securing it to the airframe structure. It shall provide two pins at the rear and positive self-locking thumbscrew fasteners at the front for solidly securing the RT-1150/ARC-159(V) to the mount. The mount with the RT-1150/ARC-159(V) attached, shall meet the crash safety requirements of MIL-E-5400 and the vibration requirements of Curve III in MIL-E-5400.

3.5.12.1.1 Form Factor - The mount shall be not greater than 9.0 inches by 5.4 inches by 1.50 inches (including flanges and thumbscrew fasteners) as shown in EI-603.

3.5.12.1.2 Weight - The weight of the mount shall be not greater than 1.8 pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.12.2 Mounting Base, Electrical Equipment MT-4760/ARC-159(V) - The mounting shall be designed for attachment to the RT-1150/ARC-159(V) in a manner equivalent to that of the MT-4658/ARC-159(V). This mount shall incorporate vibration isolators for protection of the RT-1150/ARC-159(V) from vibration. The mount, with the RT-1150/ARC-159(V) attached, shall meet the crash safety requirements of MIL-E-5400 and shall provide vibration isolation to the RT-1150/ARC-159(V) as necessary for compliance with vibration per Curve IV of MIL-E-5400.

3.5.12.2.1 Form Factor - The mount, under static conditions, shall be not greater than 11.0 inches by 7.0 inches by 2.7 inches (including flanges and thumbscrew fasteners) as shown in EI-603. Sway space for the RT-1150/ARC-159(V) on the mount shall be not greater than 11.0 inches by 7.0 inches as shown in EI-603.

3.5.12.2.2 Weight - The weight of the mount shall be not greater than 2.0 pounds.

3.5.12.3 Mounting Base, Electrical Equipment MT-4966/ARC-159(V) - The MT-4966/ARC-159(V) shall have provisions for securing the RT-1150/ARC-159(V) solidly to the aircraft comparable to those of the MT-4658/ARC-159(V). This mounting shall have additional provisions for attaching and operating the HD-1001/ARC-159A(V) for cooling the RT-1150/ARC-159(V). The cooling air shall be directed toward the rear of the receiver-transmitter and then controlled by a shroud to flow along the sides and top surfaces. The mounting shall also contain connection and switching provisions for the primary power input to the HD-1001/ARC-159(V). The HD-1001/ARC-159A(V) shall be automatically switched on and off with the receiver-transmitter on-off switching. The mounting with the RT-1150/ARC-159(V) and HD-1001/ARC-159A(V) shall have the same vibration capability as that required for the AN/ARC-159(V)2 system using the MT-4658/ARC-159(V).

3.5.12.3.1 Form Factor - The mount shall be not greater than 11.6 inches by 6.8 inches by 6.25 inches (including flanges and thumbscrew fasteners) as shown in EI-603.

3.5.12.3.2 Weight - The weight of the mount shall be not greater than 3.0 pounds.

3.5.12.3.3 Electrical Connections - The mount shall include electrical connections for attaching the HD-1001/ARC-159A(V) connector and for primary power connection to the aircraft cabling as follows and as shown in EI-603.

Reference	Receptacle Type (or approved equivalent)	Function
J1	MS3112E10 - 6P	Aircraft Power
J2	MS3112E10 - 6S	HD-1001/ARC-159(V) Power

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.12.4 Mounting Base, Electrical Equipment MT-4967/ARC-159(V) - The mounting shall be designed for attachment of the RT-1150/ARC-159(V) and HD-1001/ARC-159A(V) in a manner equivalent to that of the MT-4966/ARC-159(V). This mount shall incorporate vibration isolators for protection of the RT-1150/ARC-159(V) and HD-1001/ARC-159A(V) from vibration. The mount with the RT-1150/ARC-159(V) and HD-1001/ARC-159A(V) attached shall meet the crash safety requirements of MIL-E-5400 and shall provide vibration isolation to the mounted units as necessary for compliance with vibration per Curve IV of MIL-E-5400.

3.5.12.4.1 Form Factor - The mount, under static conditions, shall be not greater than 11.6 inches by 7.5 inches by 7.4 inches (including flanges and thumbscrew fasteners) as shown in EI-603. Overall dimensions, including sway space, of the mounting with the RT-1150/ARC-159(V) and HD-1001/ARC-159A(V) shall be not greater than 14.7 inches by 7.5 inches by 7.5 inches.

3.5.12.4.2 Weight - The weight of the mounting shall be not greater than 3.5 pounds.

3.5.12.4.3 Electrical Connections - The mounting shall have electrical connections for attaching the HD-1001/ARC-159(V) connector and for primary power connections to the aircraft cabling as follows and as shown in EI-603.

Reference	Receptacle Type (or approved equivalent)	Function
J1	MS3112E10-6P	Aircraft Power
J2	MS3112E10-6S	HD-1001/ARC-159A(V) Power

3.5.13 Mounting Base, Electrical Equipment MT-4659/ARC-159(V) - The mounting base shall be designed to solidly secure the RT-1150/ARC-159(V) receiver-transmitter and PP-7095/ARC-159(V) power supply to either MT-822/ARC-27 or MT-822A/ARC-127 in the same manner that the RT-178A/ARC-27 is secured. It shall have pins and fasteners as used on the MT-4658/ARC-159(V) and shall not require vibration isolators. The mount shall meet the crash safety requirements of MIL-E-5400.

3.5.13.1 Form Factor - The mount shall be not greater than 25.0 inches by 10.125 inches by 11.75 inches as shown in EI-603. The outer extremities, including sway space, of the combined MT-4659/ARC-159(V) and RT-1150/ARC-159(V), for assurance of physical interchange compatibility, shall be not greater than the dimensional limits of the RT-178A/ARC-27.

3.5.13.2 Weight - The weight of the mount shall be not greater than 4.75 pounds.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.13.3 Electrical Connections - The mount shall include electrical connections for direct physical and electrical connection of the RT-1150/ARC-159(V) and PP-7095/ARC-159(V) to existing aircraft wiring for the RT-178A/ARC-27 on receptacles as follows and shown in EI-603.

Reference	Receptacle Type (or approved equivalent)	Function
P1401	UG-160D/U (or equiv.)	RF Antenna
P1403	MS3102-25-7P (or equiv.)	Main/Power
P1405	MS3102-24-5P (or equiv.)	Control

3.5.14 Mounting Base, Electrical Equipment MT-4660/ARC-159(V) - The mounting base shall be designed to solidly secure the RT-1150/ARC-159(V) receiver-transmitter and PP-7095/ARC-159(V) power supply to the MT-1477/ARC-52 in the same manner that either the RT-332B/ARC-52 or RT-424A/ARC-52X is secured. It shall have pins and fasteners as used on the MT-4658/ARC-159(V) and shall not require vibration isolators. The mount shall meet the crash safety requirements of MIL-E-5400.

3.5.14.1 Form Factor - The mount shall be not greater than 20.625 inches by 10.250 inches by 8.250 inches (including connectors and fasteners) as shown in EI-603. The outer extremeities, including sway space, of the combined MT-4660/ARC-159(V) and the RT-1150/ARC-159(V) for assurance of physical interchange compatibility, shall be not greater than the dimensional limits of the RT-332B/ARC-52 or RT-424A/ARC-52X installation as shown in EI-288, Figure 2.

3.5.14.2 Weight - The weight of the mount shall be not greater than 5.0 pounds.

3.5.14.3 Electrical Connections - The mount shall include electrical connections for direct physical and electrical connection of the RT-1150/ARC-159(V) and PP-7095/ARC-159(V) to existing aircraft wiring for the RT-332B/ARC-52 or RT-424A/ARC-52X on receptacles as follows and shown in EI-603.

Reference	Receptacle Type (or approved equivalent)	Function
J1401	Amphenel 165-45 (or equiv.)	Main/Power
J1402	UG-160D/U (or equiv.)	RF Antenna

3.5.15 Mounting Bases, MT-4661/ARC-159(V) and MT-4661A/ARC-159(V) - These mounting bases shall be designed to solidly secure the RT-1150/ARC-159(V) receiver-transmitter and PP-7095/ARC-159(V) power supply to the MT-2653/ARC-51 in the same manner that the RT-780B/ARC-51AX is secured. They shall have pins and fasteners comparable to those of the

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the current version before use.

MT-4658/ARC-159(V) for the RT-1150/ARC-159(V). The MT-4661A/ARC-159(V) shall also include the HD-1001/ARC-159A(V) and have cooling accommodations for the RT-1150/ARC-159(V) comparable to those of the MT-4966/ARC-159(V). The mounts shall meet the crash safety requirements of MIL-E-5400 and shall provide vibration isolation for the RT-1150/ARC-159(V), PP-7095/ARC-159(V), and HD-1001/ARC-159A(V) on the MT-4661A/ARC-159(V), when tested to the requirements of Curve IV of MIL-E-5400.

3.5.15.1 Mounting Base MT-4661/ARC-159(V) - The following paragraphs define the MT-4661/ARC-159(V) characteristics.

3.5.15.1.1 Form Factor - The mount, under static condition, shall not exceed 14.5 inches by 9.6 inches by 6.875 inches (including connecotrs and fasteners) as shown in EI-603. Sway space for the RT-1150/ARC-159(V) on the mount shall be not greater than 15.125 inches by 9.5 inches by 7.190 inches as shown in EI-603. The outer extremities of the combined MT-4661/ARC-159(V) and the RT-1150/ARC-159(V), for assurance of physical interchange compatibility, shall be not greater than the dimensional limits of the RT-742C/ARC-51BX, RT-743B/ARC-51A, RT-767/ARC-51B or RT-780B/ARC-51AX, installation as shown in EI-436B.

3.5.15.1.2 Weight - The weight of the mount shall be not greater than 5.0 pounds.

3.5.15.1.3 Electrical Connections - The mount shall include electrical connections for direct physical and electrical connection of the RT-1150/ARC-159(V) and PP-7095/ARC-159(V) to the existing aircraft wiring for the RT-742C/ARC-51BX, RT-743/ARC-51A, RT-767/ARC-51B or RT-780/ARC-51AX on receptacles as follows and shown in EI-603.

Reference	Receptacle Type (or approved equivalent)	Function
J11	MS3112-16-26PW (or equiv.)	Control
J12	MS3112-16-26P (or equiv.)	Main/Power
J13	UG-160D/U (or equiv.)	RF Antenna

3.5.15.2 Mounting Base MT-4661A/ARC-159(V) - The following paragraphs define the MT-4661A/ARC-159(V) characteristics.

3.5.15.2.1 Form Factor - The mount under static conditions shall be not greater than 17.5 inches by 10 inches by 6.85 inches (including connectors and fasteners) as shown in EI-603. Sway space for the RT-1150/ARC-159(V) and HD-1001/ARC-159A(V) shall be not greater than 17.50 inches by 10.0 inches by 7.19 inches as shown in EI-603. The outer extremity dimensions shall be not greater than those of the AN/ARC-51 units listed in 3.5.15.1.1.

3.5.15.2.2 Weight - The weight of the mount, including the HD-1001/ARC-159A(V) shall be not greater than 6.6 pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.15.2.3 Electrical Connections - The mount shall include electrical connections for direct physical and electrical connection of the RT-1150/ARC-159(V), PP-7095/ARC-159(V), and HD-1001/ARC-159A(V) to the aircraft wiring for the AN/ARC-51 units listed in 3.5.15.1.3 as follows and as shown in EI-603.

Reference	Receptacle Type (or approved equivalent)	Function
J11	MS3112-16-26PW (or equiv.)	Control
J12	MS3112-16-26P (or equiv.)	Main/Power
J13	UG-160D/U (or equiv.)	RF Antenna

The mount shall contain electrical switching for power on-off control simultaneous with the RT-1150/ARC-159(V) power on-off control.

3.5.16 Mounting Base, Electrical Equipment MT-4664/ARC-159(V) - The mounting base shall be designed to solidly secure the RT-1150/ARC-159(V) receiver-transmitter and PP-7095/ARC-159(V) power supply to the MT-1099( )/U in the same manner that the RT-463A/ARC-34B is secured. It shall have pins and fasteners as used on the MT-4658/ARC-159(V) and shall not require vibration isolators. The mount shall meet the crash safety requirements of MIL-E-5400.

3.5.16.1 Form Factor - The mount shall be not greater than 21.375 inches by 10.125 inches by 7.600 inches. The outer extremities of the combined MT-4664/ARC-159(V) and RT-1150/ARC-159(V), for assurance of physical interchange compatibility, shall be not greater than the dimensional limits of the RT-463A/ARC-34B.

3.5.16.2 Weight - The weight of the mount shall be not greater than 4.0 pounds.

3.5.16.3 Electrical Connections - The mount shall include electrical connections for direct physical and electrical connection of the RT-1150/ARC-159(V) and PP-7095/ARC-159(V) to existing aircraft wiring for the RT-463A/ARC-34B as follows and shown in EI-603.

Reference Designation	Receptacle Type (or approved equivalent)	Function
J1101	Winchester No. QRE34P (or equiv.)	Main/Power
P1111	UG-160D/U (or equiv.)	RF/Antenna

3.5.17 Mounting Base for RT-1194/ARC-159A(V) - The mounting shall provide for mounting the RT-1194/ARC-159A(V) to the aircraft, and for mounting of several AN/ARC-159A(V) ancilliary units if they are included in the installation. These ancilliary units are as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

a. HD-1001/ARC-159A(V) Cooler, Air, Electronic Equipment
b. PP-7323/ARC-159A(V) Power Supply
c. F-1420/ARC-159A(V) Filter-Converter

Requirements of the RT-1194/ARC-159A(V) mountings and these ancillary units are defined in the following sections:

3.5.17.1 Mounting Base Electrical Equipment MT-4795/ARC-159A(V) - The mounting base shall be designed to secure the RT-1194/ARC-159A(V) receiver-transmitter to the MT-2653/ARC-51 in the same manner that the RT-742C/ARC-15BX ,RT-743B/ARC-51A or RT-780B/ARC-51AX is secured. It shall have pins and fasteners for RT-1194/ARC-159A(V) attachment that permit easy removal of the receiver-transmitter from the mounting. The mount with the receiver-transmitter attached shall meet the crash safety requirements of MIL-E-5400. The mount and receiver-transmitter combination shall withstand failure vibration per MIL-E-5400 Curve IV limits.

3.5.17.1.1 Form Factor - The mount, under static conditions, shall be not greater than 9.875 inches wide by 6.9 inches high by 15.625 inches long (including connectors and fasteners) as shown in EI-603. Sway space for the RT-1194/ARC-159A(V) and mounting shall be not greater than 10.125 inches wide by 7.187 inches high by 17.5 inches ling, as shown in EI-603. The outer extremities of the combined MT-4795/ARC-159A(V) and the RT-1194/ARC-159A(V), for assurance of physical interchange compatibility shall be not greater than the dimensional limits of the RT-742C/ARC-51BX, RT-743B/ARC-51A or RT-780B/ARC-51AX.

3.5.17.1.2 Weight - The weight of the mount shall be not greater than 5.3 pounds.

3.5.17.1.3 Electrical Connections - The mount shall include electrical connections for direct physical and electrical connection of the RT-1194/ARC-159A(V) to existing wiring for the RT-742C/ARC-51BX, RT-743B/ARC-51A, or RT-780B/ARC-51AX on receptacles as follows and as shown in EI-603. (The +28 volt DC primary power input available must be compatible with the RT-1194/ARC-159(V) primary power requirements.)

Reference Designation	Receptacle Type (or approved equivalent)	Function
J11	MS-3112-16-26PW (or approved equiv.)	Control
J12	MS-3112-16-26P (or approved equiv.)	Main/Power

3.5.17.1.4 Cooling - The MT-4795/ARC-159A(V) shall have provisions at the rear for attaching a duct for aircraft supplied cooling air to the RT-1194/ARC-159A(V) shown in EI-603.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.17.1.5 Provisions for Ancillary Units - The MT-4795/ARC-159A(V) shall have the following provisions for ancillary units.

a. HD-1001/ARC-159A(V) Cooler, Air, Electronic Equipment - The air duct provisions defined in 3.5.17.1.4 shall be suitable for use and mounting of the HD-1001/ARC-159A(V). A connector shall be available on the mount for supplying 115V AC power to the cooler.

b. PP-7323/ARC-159A(V) Power Supply - The mounting shall have securing provisions, connector and wiring accommodations for the PP-7323/ARC-159A(V) converter, electrical equipment. This converter may or may not be used, depending on the primary power availability in the aircraft.

c. F-1420/ARC-159A(V) Filter-Converter - The mounting shall provide for the securing and electrical connections to the F-1420/ARC-159A(V). This filter converter shall be located beside the RT-1194/ARC-159A(V) and shall have mounting provisions comparable to those of the receiver-transmitter.

3.5.17.1.6 Internal Wiring and Circuitry - The MT-4795/ARC-159A(V) shall have connectors to aircraft cabling as listed in 3.5.16.1.3 plus the connectors, wiring and circuits to accommodate the RT-1194/ARC-159A(V), and units listed in 3.5.16.1.5. The mounting shall also include electronic circuitry for amplification of the voice audio input modulation of 0.5 volts RMS to 4.2 volts RMS for usage during secure voice communication. The mounting shall have jumper or equivalent switching provisions for direct usage of the 4.2 volt secure voice input when such is available at the input connector.

3.5.18 Mounting Base, Electrical Equipment MT-4839/ARC-159A(V) - The mounting shall be designed for attachment of the RT-1194/ARC-159A(V), to the aircraft. The mount shall have provisions for attachment of the 3.5.10.

The mount with the RT-1194/ARC-159A(V) attached shall meet the crash safety requirements of MIL-E-5400 and shall provide vibration isolation of the RT-1194/ARC-159A(V) as necessary for compliance with vibration per Curve IV of MIL-E-5400.

3.5.18.1 Form Factor - The mount, under static conditions, shall be not greater than 15.2 inches by 7.5 inches by 7 inches (including flanges and thumbscrew fasteners) as shown in EI-603.

3.5.18.2 Weight - The weight of the mount shall be not greater than 3.0 pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.19 Mounting Base, Electrical Equipment MT-4840/ARC-159A(V) - This mounting shall be designed for attachment of the RT-1194/ARC-159A(V) and the F-1420/ARC-159A(V) to the aircraft. Securing provisions for the two mounted units shall be independent allowing independent removal of either unit. The mount shall have provisions for attachment of the 3.5.10 at the rear of the 3.5.3.2.

The mount, with the RT-1194/ARC-159A(V) and the F-1420/ARC-159A(V) attached shall meet the crash safety requirements of MIL-E-5400 and provide vibration isolation of the RT-1194/ARC-159A(V) and the F-1420/ARC-159A(V) as necessary for compliance with vibration per Curve IV of MIL-E-5400.

3.5.19.1 Form Factor - The mount, under static conditions, shall be not greater than 15.2 inches long by 10.125 inches wide by 7.0 inches high (including flanges and thumbscrew fasteners) as shown in EI-603.

3.5.19.2 Weight - The weight of the mount shall be not greater than 3.5 pounds.

3.5.20 Mounting Base, Electrical Equipment MT-6093/ARC-159(V) - This mounting base shall be designed for solidly securing the AN/ARC-159(V)1 to aircraft or small boats or craft structures. It shall also include interface wiring, switching and circuitry for system compatibility with the C-10206/URC-94. Physically, the mounting shall have screw holes in the base for use in attachment to the aircraft or boat. It shall also have rail fastener provisions for securing the AN/ARC-159(V)1 with its front panel fasteners. The configuration shall form a protective enclosure around the sides and top of the AN/ARC-159(V)1.

3.5.20.1 Connectors and Controls - Power and audio connectors for connection to external equipment shall be located in the base at the rear of the MT-6093/ARC-159(V). Toggle switches for panel light control and secure voice equipment mode control shall be located in the base at the front.

3.5.20.2 Environmental Specifications - The mounts with the AN/ARC-159(V)1 attached shall meet the crash safety requirements of MIL-E-5400. The mount and radio set combination shall withstand without failure vibration per MIL-E-5400 Curve III limits.

3.5.20.3 AN/ARC-159(V)10 System - The MT-6093/ARC-159(V) with the AN/ARC-159(V)1 radio set installed form the AN/ARC-159(V)10 system.

3.5.20.4 Form Factor - The mount overall dimensions shall be not greater than 9.8 inches in lenght, 8.75 inches in width and 7.5 inches in height. When installed, the mating connectors will project beyond this length in the rear and the AN/ARC-159(V)1 knobs may project up to 1.40 inches in the front. Detail configuration information and installation parameters are defined in EI-603.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the current version before use.

3.5.20.5 Weight - The weight of the MT-6093/ARC-159(V) shall be not greater than 5.75 pounds.

3.5.20.6 Electrical Connections - The mount shall include electrical connectors and cabling, except for the antenna connector, for direct connection to the AN/ARC-159(V)1 rear connectors. The cabling shall be connected to the two connectors, power and audio, at the rear of the MT-6093/ARC-159(V) for installation interconnect wiring. These connections and reference designations are as follows:

Reference	Receptacle Type (or approved equivalent)	Function
J1	MS27508E-12-4P	Power
J2	MS27508E-14-18S	Audio

3.5.20.7 Internal Wiring and Circuitry - In addition to the direct interconnect wiring from the AN/ARC-159(V)1 to the rear connectors, the MT-6093/ARC-159(V) shall contain a microphone amplifier circuits. This circuit shall amplify the 0.25 volt RMS audio from the C-10206/URC-94 to a level of 0.5 volts RMS ±10 percent at 150 ohms impedance for the AN/ARC-159(V)1 modulator input. The mounting shall also include a resistor attenuations of the AN/ARC-159(V)1 unattenuated audio to provide 6 dB ± 1.5 dB attenuation when externally terminated by 600 ohms.

3.5.20.7.1 Switch Functions - Two toggle switches at the front of the mounting shall provide the following controls: The panel lamp intensity toggle switch shall have three positions for off-low intensity-high intensity for panel lamp illumination intensity control. The second toggle switch shall have two positions for secure voice equipment baseband-diphase mode control, which in turn serves as the AN/ARC-159(V)1 narrowband and wideband control. The switch functions and positions shall be appropriately labeled on the mounting structure.

3.5.20.7.2 Fuse Requirement - The MT-6093/ARC-159(V) shall have a protective fuse at the input of the 27.5 volt primary power line. It shall also contain a relay for providing 27.5 volt power to any external system units when the AN/ARC-159(V)1 radio set is turned on.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:37Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract, the contractor is responsible for the performance of the inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein. The Government reserves the right to perform any of the tests set forth in the specification where such tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Inspections - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

a. Preproduction (First Article) Tests (see 4.2)
b. Initial Production Tests (see 4.3)
c. Acceptance Tests (see 4.4)

4.2 Preproduction (First Article) Tests - When specified in the contract, preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Tests shall be conducted and test data collected per the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government. Data collected in conducting the tests shall be included.

4.2.1 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Preproduction tests shall include environmental tests in accordance with MIL-T-5422, except that the watertightness test shall be in accordance with MIL-STD-810 Method 506.

4.2.2 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

4.2.3 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government and it shall be the obligation of the contractor to make any necessary corrections.

4.3 Initial Production Tests - As listed in 6.2.1 and when specified in the procurement document, one of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also include such accessory material and operational information necessary to test the equipment. (See 6.2.1)

4.3.3 Initial Production Sample Test Results - Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all limited life or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall prepare test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall include the compiling of test data and shall consist of the following:

a. Individual Tests (see 4.4.1)
b. Reliability Assurance Tests (see 4.4.2)
c. Random Vibration Tests (see 4.4.3)

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of Product (see 4.4.1.1)
b. Operational Test (see 4.4.1.2)
c. Manufacturing Run-in Test (see 4.4.1.3)

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run-In (Burn-In) Test - All units identified in 3.5 Detail Requirements except mountings blowers and secure voice control units shall be tested for a period of 60 hours - the last 18 failure-free-in accordance with MIL-STD-781 under the following conditions. The temperature cycling shall consist of 4½ hour cycles, each cycle consisting of approximately 1½ hours off at -54°C and 3 hours operating during the +71°C, +65°C, and +55°C temperature sequence as illustrated in Figure 1. The equipment shall be subjected to 2.2 Gs ± 10 percent vibration at 30 Hz for 10 minutes of each hour of operating time. Prior to the start of the burn-in test the equipment performance shall be measured at -54°C and +55°C. Measurement of major parameters is required. For example, the receiver transmitter shall be checked for sensitivity, audio output, frequency accuracy, RF output and modulation levels. Indicators shall be checked for appropriate readout. The above characteristics shall also be measured once during each normal working day. Performance monitors shall be installed or continuous monitoring of the equipment overall performance, transmitter power output and receiver sensitivity, during the test period, for detection of intermittent conditions. The test results of this test shall be available for procuring activity review at the contractor's facility.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

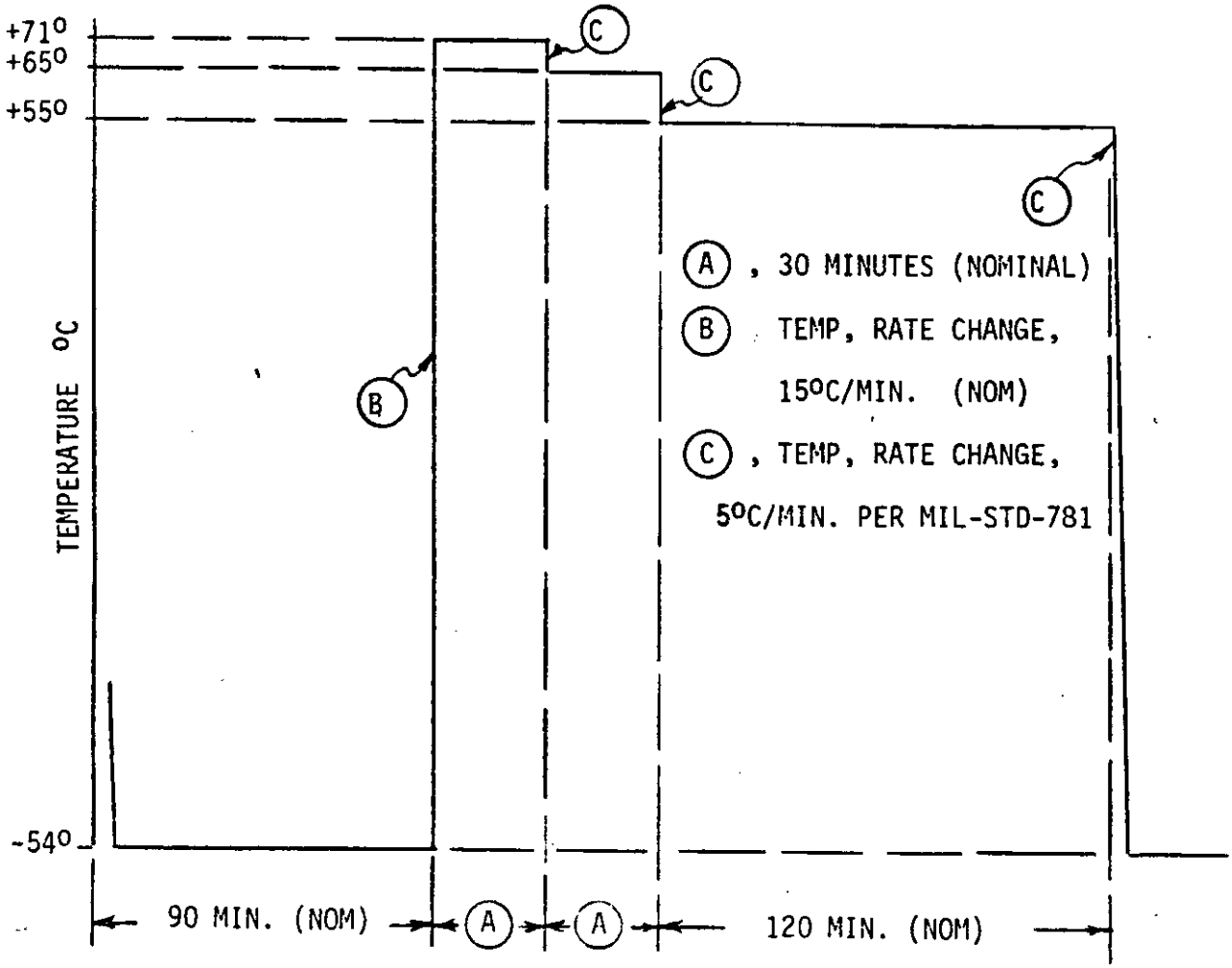


FIGURE 1. Temperature-time profile for manufacturing run-in (burn-in) test temperature cycling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

4.4.1.3.1 Spare Module Tests - Burn-in tests shall be applicable to spare module procurements. Burn-in requirements for spare parts and subassemblies shall be as specified in the contract. (See 6.2.1)

4.4.2 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted in accordance with MIL-STD-781. Classification of failure shall be in accordance with MIL-STD-781 and MIL-STD-2704. Tests shall be performed on radio sets or receiver-transmitter and control unit combinations, with an appropriate indicator if included in the procurement: Reliability requirements for any units or combination of units shall be as specified in 3.3.2.3.

4.4.2.1 Qualification Phase Tests - When specified in the contract, (see 6.2.1), and prior to the acceptance of equipments under the contract, a minimum of three (3) equipments shall be tested in accordance with MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, test level E of MIL-STD-781 shall be used. The Accept-Reject Criteria for Test Plan IV shall be used.

4.4.2.2 Reliability Production Acceptance (Sampling) Phase Tests - (See 6.2.1) When specified in the contract, the equipment, throughout production, shall be tested in accordance with MIL-STD-781 (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests". Test level E shall be used.

4.4.2.2.1 Lot Size for Sampling Phase - Each monthly production lot shall be subjected to a Reliability Sample Test in accordance with MIL-STD-781. The sample size shall be as follows:

Monthly Production Quantity	Sample Size
2 to 5	1
6 to 25	2
26 to 50	3
51 to 100	4
101 to 200	5
201 and over	6

The test duration shall be a minimum of 256 hours per each unit tested. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

(Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month and as listed in 6.2.2 the test results shall be made available to the procuring activity. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be informed. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.3 Random Vibration Tests - When specified in the contract, (see 6.2.1) additional run-in testing on a sampling basis shall be conducted on the panel mounted radio sets. These additional tests shall include random vibration of 4.4 G rms, 15 Hz to 2000 Hz for 10 minutes in each major plane. The receiver and transmitter major functions shall be monitored during the vibration test. The sampling quantity and test schedule shall be as stated in the contract. (See 6.2.1)

4.4.4 Special Tests - When specified in the contract, (see 6.2.1) special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the reliability assurance tests.

4.4.4.1 Scope of Tests - Special tests shall consist of such tests as acceptable to the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable.

4.4.5 Equipment Failure - Should a failure occur during either the reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or design defect.

c. Identify for the procuring activity the proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

4.6 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor for procuring activity approval. (See 6.2.2) The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures may be used without further approval. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmission, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be made available to the Government inspector.

5. PACKAGING

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended Use - The AN/ARC-159 and AN/ARC-159(V) radio sets shall be of a variable configuration using common modules that can be used to meet the prime or backup UHF AM voice communication requirement in any tactical aircraft. The variable configurations are intended to provide direct physical replacement of other radio sets (i.e. AN/ARC-27, AN/ARC-34, AN/ARC-51, and AN/ARC-52) without requiring revision of the existing aircraft wiring and mounting provisions. They are intended for production incorporation in all new aircraft plus retrofit incorporation in selected out of production aircraft.

6.1.1 AN/ARC-159 Applications - The AN/ARC-159 radio set is intended for applications requiring a console mounted radio with capabilities as described in 3.5.1.1. It is not intended for applications requiring direct physical replacement of other radio sets (i.e. AN/ARC-27, AN/ARC-34, AN/ARC-51, and AN/ARC-52) or provisions for remote control and remote frequency-channel indication. It is not intended to have the capabilities for guard channel precedence operation or transmission of CASS/DICASS signals.

6.1.2 AN/ARC-159(V) Applications - The AN/ARC-159(V) is a designation for a variable configuration radio set. It is intended for applications requiring either a console mounted radio with capabilities as described in 3.5.2.1, or a remote controlled radio set with capabilities as described in 3.5.3 and 3.5.4. The designation of the variable configuration radio sets and the basic component equipments of each is as follows:

Designation	Component Equipments	Radio Set Replaced
AN/ARC-159(V)1	AN/ARC-159(V)1	AN/ARC-159
AN/ARC-159(V)2	RT-1150/ARC-159(V) C-9577/ARC-159(V) MT-4658/ARC-159(V)	None
AN/ARC-159(V)3	RT-1150/ARC-159(V) C-9951/ARC-159(V) MT-4659/ARC-159(V)	AN/ARC-27
AN/ARC-159(V)4	RT-1150/ARC-159(V) C-9952/ARC-159(V) MT-4660/ARC-159(V)	AN/ARC-52
AN/ARC-159(V)5	RT-1150/ARC-159(V) C-9946/ARC-159(V) MT-4661/ARC-159(V)	AN/ARC-51
AN/ARC-159A(V)5	RT-1194/ARC-159A(V) C-9946/ARC-159(V) MT-4795/ARC-159A(V)	AN/ARC-51

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

Designation	Component Equipments	Radio Set Replaced
AN/ARC-159(V)8	RT-1150/ARC-159(V) C-9954/ARC-159(V) MT-4664/ARC-159(V)	AN/ARC-34
AN/ARC-159(V)10	RT-1150/ARC-159(V) MT-6093/ARC-159(V) *C-10206/URC-94	

*Not considered as an AN/ARC-159(V) unit.

In addition to the basic component equipments, each variable configuration listed above may be used with the following equipments as required for their specific applications:

a. An additional control may be used with any AN/ARC-159(V)1 through AN/ARC-159(V)8 configurations for remote control applications.

b. One or two indicators ID-1972/ARC-159(V) or ID-1984/ARC-159(V) or OD-122/ARC-159(V) may be used with any AN/ARC-159(V)1 through AN/ARC-159(V)8 configurations for remote frequency/channel indications.

c. The switching unit SA-1964/ARC-159(V)1, control C-9451/ARC-159(V), and mount MT-4609/ARC-159(V) may be used with two AN/ARC-159 or any AN/ARC-159(V) configuration for TSEC/KY-28 control and automatic relaying operations.

The switching unit SA-2078/ARC-159(V)1, control C-9960/ARC-159(V), and mount MT-4609/ARC-159(V) may be used with two AN/ARC-159 or any AN/ARC-159(V) configuration and two TSEC/KY-28 equipments for independent system plain or secure voice operation or for automatic relaying operation.

d. The C-9451/ARC-159(V) may be used as a substitute for the C-8057/ARC for control of the TSEC/KY-28.

e. With appropriate connector and wiring accommodations, the smaller control unit series 3.5.4.2 may be used with any AN/ARC-159(V)1 through AN/ARC-159(V)8 system.

6.2 Ordering Data -

6.2.1 Procurement Requirements - The procurement document should specify the following:

a. Title, number and date of this specification
b. Equipment title and type designation

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the currentversion before use.

c. Panel lighting color (red or white)
d. Selection of applicable levels of packing or packing.

The following test and inspection options, if exercised, must be stated as required in the contract:

a. Preproduction (First Article) Tests (see 4.2)
b. Initial Production Tests (see 4.3)
c. Qualification Phase Tests (see 4.4.2.1)
d. Reliability Production Acceptance (Sampling) Phase Tests (see 4.4.2.2)
e. Random Vibration (see 4.4.3) The sampling quantity and test schedule shall be identified.
f. Special Tests (see 4.4.4)
g. Spare parts and subassembly burn-in tests to be performed on spare procurements. (see 4.4.1.3.1)
h. Reliability Program on reorder, if other than program previously used (see 3.3.2)

(External cables and that portion of the connectors attached to the cables are not supplied as part of the equipment.)

6.2.2 Data Requirements - When this specification is used in a procurement which incorporates a DD-1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs.

a. Applicable for Reordered Equipments

Paragraph	Data Requirement	Applicable DID
4.6	Test Procedures	UDI-T-21347

When approved test procedures from previous contracts are available, they may be considered applicable without further procuring activity approval.

b. Additional Data if Preproduction Tests Required

Paragraph	Data Requirement	Applicable DID
4.3.2	Test Report	UDI-T-21349
4.6	Test Procedures	UDI-T-21347

c. Additional Data if Initial Production Tests Required

Paragraph	Data Requirement	Applicable DID
4.3.2	Accessory Material	UDI-E-21338

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

d. Additional Data if Qualification Phase Tests Required

Paragraph	Data Requirement	Applicable DID
4.6	Test Procedures	DI-R-7035
4.4.2.1	Test Report	DI-R-7034

e. Additional Data if Reliability Acceptance (Sampling) Phase Tests Required

Paragraph	Data Requirement	Applicable DID
4.6	Test Procedures	DI-R-7035
4.4.2.2	Test Reports	DI-R-7034

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Instructions for First Article - When a first article is required for inspection and approval (see 3.1, 4.2, and 6.2.1).

6.4 Precedence of Documents - When the requirements of the contract, this specification, of applicable subsidiary specifications are in conflict, the following precedence shall apply:

a. Contract - The contract shall have precedence over any specification.

b. This Specification - This Specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

c. Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.5 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation and information compiled shows advantages can be realized by a design change or use of different materials, it is recommended the findings be made available for procuring activity review.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

6.6 Type Designations - The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Changes From Previous Issue - Asterisks are not used in this issue to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.8 Associated Equipment - This equipment shall operate with the following associated equipment which shall not be supplied as part of this equipment:

Item	Equipment Designation
Intercommunication Sets	AN/AIC-10, AN/AIC-14 AN/AIC-18 or AN/AIC-25
Headset-Microphone	H-157/AIC
Headset	H-173/AIC
Microphone, Boom Mounted	M-96A/A
Amplifier, Audio Frequency	AM-3597B/A
Control	C-824A/AIC-10, C-2106/AIC-18, C-6567/AIC-25, or C-6533/AIC
Antenna	AT-141A/ARC or AT-256A/ARC
Switch	SA-521/A with Selector, Antenna C-2193B/A
Direction Finder Group	AN/ARC-25, AN/ARA-48 or AN/ARA-50
Speech Security Equipment	TSEC/KY-28 or equivalent
Reference Signal Generator	Unit 1 and Unit 2 of AN/ASA-76

6.9 Metric Equivalents - Dimensions and weights of units included herein are listed in inches and pounds respectively. For conversion to metric system equivalents, the following multiplier of factors are applicable:

Pounds to Kilograms - Multiply by 0.454
Inches to Centimeters - Multiply by 2.540

Preparing Activity
NAVY AS

(Project 5821-N149)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Specifications And Standards Dept. (ESSD)
Code 93
Lakehurst, New Jersey 08733

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-R-81877(AS) Radio Sets, AN/ARC-159 and AN/ARC-159(V)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

☆ U.S. GOVERNMENT PRINTING OFFICE: 1973—714-866/3983 2-1

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the current version before use.